

# Pan-CoV epitope sterically occluded in an open early spike fusion intermediate

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**This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.**

Version 0:

Reviewer comments:

Referee #1

(Remarks to the Author)

This work studies antibodies against the S2' helix, an area of interest as it is conserved across coronaviruses. This epitope is exposed after ACE2 priming. An interesting question is why does the neutralization capacity of these antibodies of interest diminish in Omicron given that Omicron contains no mutations in the epitope? Another remaining question is how antibodies against this epitope achieve their neutralization – presumably blocking of conformational change – but exactly how is unknown. At the end of the day, these authors find through binding studies and cryoEM, that Omicron mutations elsewhere decrease access to this epitope, and that Omicron further evades these antibodies by switching to a cathepsin-mediated entry pathway. Notable structural biology studies include elucidation of intermediates in the refolding pathway of spike- I see the early intermediate structures, but not the late ones. Did I miss them in the paper? Some other observations are that Omicron seems to refold faster than WT, and that 76E1 cant completely inhibit refolding to postfusion in Omicron, although it could in WT.

Their findings suggest that scFv will more effectively target this site than full antibodies, and that accessibility of the site to the proteases might also influence infectivity.

I'm not fully understanding the experiments of 1b and 1d, but it is clear that scFv neutralizes better than Fab which does suggest an epitope access issue. scFv neuts better than Fab against other coronaviruses as well.

One of the Omicron mutations H655Y reduces neut by these antibodies – the authors find that it must limit epitope exposure as scFv neutralizes H655Y-bearing spike more than Fab.

They analyze contacts in previous structures and suggest that this mutation may alter the size of the shift or propensity of S1 to shift from S2 which alters epitope exposure.

Comparison to crystal structures suggests that the orientation of the different antibodies influences their ability to bind and neutralize. Further. Omicron uses cathepsin mediated entry in the endosome.

I have focused on big picture here rather than experimental detail. Any exploration of the spike epitope conserved across coronaviruses and how to make recognition of these epitopes more successful is good. The finding that they are compromised in access and some insight into why will also guide some eventual therapeutic utility. Of chief importance is that fusion intermediate structure, although I wish I had seen a structure of the later intermediate, even a low resolution one, as well. I see a light blue trimer in extended data figure 3 – but this density may be three copies of the antibody and not the spike intermediate. These structures are difficult to achieve but would be notable if possible.

(Remarks on code availability)

Referee #2

(Remarks to the Author)

Building on their previous work, the authors provide structural and biochemical evidence that Omicron evades antibodies targeting the S2'-helix through two primary mechanisms: reducing S2'-helix exposure while accelerating spike refolding and shifting viral entry toward a CTSL-dependent pathway. The study presents high-resolution cryo-EM structures of spike refolding intermediates transitioning from the prefusion to postfusion state, which is valuable for understanding spike dynamics and informing future antibody design. However, some conclusions remain speculative without direct experimental support. For instance, the impact of TMPRSS2 and CTSL size, shape, and binding orientation on S2-helix accessibility and subsequent cleavage by neutralizing antibodies is not addressed. Additionally, S1 shedding in response to ACE2 treatment and the effect of antibodies on membrane fusion should be compared, given the role of ACE2 priming in S2 cleavage and fusion activation. Quantification of protein band intensities would further strengthen the conclusions. A high-resolution structure of XBB.1.5 or JN.1 spike in complex with E-FIC-76E1 would also be informative, considering similarities in 2D classification and 3D reconstruction between WT and XBB.1.5 L-FIC-76E1.

Specific comments:

- Fig. 1E and other Western blots: Actin and spike signal variations across samples necessitate the tested antibody input control. S1 shedding should be shown, and band intensities quantified. Clarification on whether WT refers to the ancestral Wuhan strain or D614G is needed.
- Neutralization and structure: Assessing COV44-62 and C77G12 binding to the SARS-CoV S2'-helix will be helpful.
- Fig. 3: It's important to quantify S and S2' band intensities and demonstrate dose-dependent effects of TMPRSS2, CTSA, and antibody treatments, with IC50 calculations for WT and BF.7. Include shed S1 as a control to compare ACE2 priming effects. The effects of E64d and Camostat appear modest, and the Delta strain, which primarily utilizes plasma membrane entry, could serve as a control.
- Acidic environment and antibody escape: It will be important to test Ab binding and shedding over time under different pH conditions at 37°C, rather than 20 min, to substantiate the claim that acidic pH does not affect on Ab shedding.
- H655Y mutation and entry pathway shift: The assertion that H655Y shifts viral entry preference to the endosomal route by decreasing S2'-helix exposure requires additional experimental validation. Reversion key mutants in Omicron should be tested for entry pathway and S2'-helix exposure.
- Comparative spike refolding exposure: Experimental evidence should support claims that SARS-CoV-1 and HCoV-229E spikes exhibit less exposed S2-helix during refolding compared to MERS-CoV and HCoV-NL63, particularly in response to full-length and scFv antibodies.

(Remarks on code availability)

Referee #3

(Remarks to the Author)

In this manuscript, the authors describe how the format/size of S2' helix antibodies modulate their neutralization potency in various context and they obtained a cryogenic-electron microscopy structure of a SARS-CoV-2 spike fusion intermediate with the 76E1 Fab. The results are interesting although not all conclusions are supported by the data and some of the results appear to overlap with an article published in Cell by some of the same authors (PMID: 39889696).

Lines 113-117

Have the authors controlled for differences of WT vs XBB spike expression?

Lines 124-130

How do the authors explain the strong scFv binding of COV44-62 and C77G12 in the absence of ACE2 (Figure 1e)?

Lines 136-139

How do the authors reconcile the conclusion that Omicron variants reduce S2' helix epitope exposure given that they were able to image a stalled E-FIC for both WT and XBB.1.5 spikes? Are the cryoEM structures suggesting an explanation for this mechanism?

Lines 149-151

Please rephrase as it is simply that 76E1 does not bind prefusion spike

Lines 155-157

How do the authors know that the reconstruction obtained correspond to the L-FIC considering the severe preferred orientation and associated low resolution ?

Lines 159-163

CH does not change conformation relative to prefusion spike

Lines 186-188

Has a control experiment been carried out without 76E1 and was post fusion observed?

The authors claim that Omicron spike refold more rapidly than WT spike. This statement is not supported by any data on refolding kinetics.

Lines 198-200

It suggests reduced stability (delta G) not faster refolding kinetics of individual spikes.

Also, how do the authors reconcile this with the presence of D614G in XBB.1.5 but not WT which is supposed to enhance prefusion stability?

Lines 201-210

What is the molecular rationale for this?

Were the native spike ECDs co-expressed with furin (as recently done by the same authors, PMID: 39889696)? If not, what was the rationale to employ a different approach?

Have the authors tried adding 76E1 scFv and did this prevent transition to the postfusion spike?

Why would 76E1 binding to XBB.1.5 spike require detachment of S1-ACE2 or further outward extension than WT? What evidence supports this statement?

Is there an Extended Data Figure describing data collection and processing of this XBB.1.5 cryoEM dataset?

Lines 226-228

Are the authors referring to differences of relative orientation between constant and variable domains (i.e elbow angle)?

Could it be that the 3 Fabs compared crystallized with distinct packing, hence the different elbow angles observed, but that they would be able to flex around it to bind to EIC (as the elbow angle is notoriously flexible for virtually all mAbs)?

Lines 241-245

Have the authors tested cleavage with TMPRSS2 found on target cells (trans) as opposed to the spike-producing cells (cis)?

Lines 252-262

TMPRSS2 is indeed known to be expressed in Caco2 cells but cathepsins are too (hence the inhibitory effect of camostat and E64d in Fig3e for BF.7 and XBB.1.5)

The authors seem to imply that a switch from a TMPRSS2-dependent to a CTSL-dependent entry pathway occurred to evade S2' helix directed antibodies. However, all reports have indicated that these types of antibodies are both rare and weakly neutralizing and it is therefore unlikely that they would be able to exert an appreciable selective pressure that would have led to this outcome.

How do the authors reconcile the proposed reduced/loss of TMPRSS2 dependence for Omicron variants (observed by several other groups) with contradictory results in organoid models (PMID: 37555660) and in vivo (PMC9858505)?

Extended Data Fig7a

Why are the binding responses (OD) so small in these ELISAs?

Why using BA.1 S instead of BF.7 in a-b?

Lines 304-307

These two sentences seem contradictory as CTSL entry will always be the route used in 293T-ACE2 cells

Lines 319-321

Have the authors imaged the ACE2/S+H655Y to confirm that S1 is not moving as much as in the structures they obtained here and in their recent Cell paper?

Lines 323-324

The data presented in Fig4f indicate that all pseudoviruses tested depend on endosomal entry only in (293T-ACE2 cells) and are affected by inhibitors of both plasma membrane (camostat) and endosomal (E64d) entry. This sentence therefore seems to be an overstatement

What evidence do the authors have to support the lower opening of S1 for XBB vs WT?

Lines 377-379

The accumulation of E-FIC should not be an issue for 76E1 binding as it is a structure that was obtained by the authors. This does not make sense and is in discrepancy with the data shown in Fig2a: 25.4% (WT) vs 15.3-20.7% (XBB.1.5) for E-FIC. Refolding to postfusion is the issue as 76E1 cannot bind to this conformation.

Based on the file names and on the appearance of the maps, it appears that the authors provided deep learning enhanced reconstructions. While the use of such maps can help model building, unmodified maps should be provided and deposited to the EMDB. This reviewer encourages the author to do the same and update the maps deposited for their recently published Cell paper.

(Remarks on code availability)

#### Referee #4

(Remarks to the Author)

The continued emergence of coronaviruses in the human population demonstrates the importance of developing vaccines and therapeutics that confer broad protection. The spike protein, responsible for receptor binding and cell fusion, has been the focus of vaccines and monoclonal antibody therapies. However, the antigenic evolution of the spike has had a significant impact on vaccine efficacy and has rendered all clinical-stage monoclonal antibodies (mAbs) ineffective. Identifying and characterizing conserved regions in the spike is crucial for developing more robust countermeasures.

One of the most promising targets within the spike is the S2'-helix, a pan-coronavirus epitope buried in the spike, which becomes transiently exposed upon ACE2 binding. While the binding of these antibodies has been characterized in the context of isolated peptides, the molecular basis for their inhibitory mechanism of action in the context of spike refolding remained unknown – in part due to the transient nature of spike fusion intermediates. Moreover, mechanistic studies into potential immune evasion of these important antibodies by SARS-CoV-2 variants were lacking.

Through structural and functional studies, Bao et al. reveal that Omicron variants evade S2'-helix-targeting antibodies by reducing epitope exposure, speeding up spike refolding, and shifting to a Cathepsin L-dependent entry pathway, with the H655Y mutation playing a crucial role. Through antibody minimization experiments, the authors demonstrate that protein size affects its access to the S2'-helix. Extensive cryoEM analysis shows that these antibodies stabilize both early and late intermediates of the SARS-CoV-2 spike, but they capture fewer early intermediates of the Omicron spike due to reduced epitope exposure and faster refolding. Notably, minimizing antibody size reversed these negative effects and significantly boosted neutralizing activity against Omicron variants and other human coronaviruses, including SARS-CoV-1 and HCoV-229E.

This impressive work provides important new information about this crucial epitope and offers tangible strategies for designing pan-coronavirus biologics by targeting the S2'-helix, paving the way for more efficient and broad therapeutic strategies. As such, I believe this work would fit within the remit of Nature once several issues have been addressed.

#### Main concern

In Figure 1A and Extended Data Figure 3, the authors quantify different conformation states of the spike protein. Based on inspection of the 2D and 3D class averages, I mostly agree with the authors' interpretation. However, the clear anisotropy for certain conformations, such as the XBB.1.5 E-FIC and L-FIC, leaves some room for interpretation. For example, how do the authors know that the L-FIC is a true late intermediate and not the product of damaged particles? Conceivably, absorption to the air-water interface could result in such classes. I think it would be wise to include such a short caveat in the main or supplementary text to cover this possibility, perhaps in lines 183-185, where the authors suggest the lack of density is due to "a dynamic characteristic." Moreover, given the heterogeneity and importance of these data sets, I would strongly advise that the authors deposit the raw data from Figure 1A and Extended Data Figure 3 to EMPIAR. The reason for this is that unlike the larger 11,263 data set, the community has no way to assess the product of the processing, which in this case is 2D and 3D class averages.

#### Minor concerns

- Related to my main concern, in lines 204-207, the authors state: "Despite similarities in 2D classification and 3D reconstruction between WT and XBB.1.5 L-FIC-76E1, attempts to obtain a high-resolution structure of XBB.1.5 E-FIC-76E1 were unsuccessful, indicating its heterogeneity." I presume the authors intend to refer to only E-FIC in this passage; otherwise, it does not make sense. Moreover, I would say that the primary reason for the inability to obtain a structure of E-FIC is due to a lack of side views and not its heterogeneity.
- In lines 169-174, the authors write the description of the epitope-paratope interactions in a manner that suggests this is all readily interpretable from the cryo-EM maps. Given the modest resolution, and the fact the maps have used density modifying software, which I am not generally in favour of, it is a bit misleading. Moreover, this is not necessary as this information is already available from the crystal structure, used as a starting model and mentioned in lines 180-181. The strength of the cryo-EM data in the present study is that it provides new insights into the quaternary structure of the complex at modest resolution. The description of the atomic details can be shortened and toned down.
- In lines 300-303, the authors state: "Similar results were found in the neutralization assay against H655Y pseudovirus in 293T-ACE2 cells, with scFv demonstrating a more significant enhancement of neutralizing activity (Fig. 4e)." To me, the phrasing of this sentence implies that they are comparing the FL to scFv enhancement in H655Y to something else, presumably the WT. Should the authors not then point to another figure to demonstrate this, such as Figure 1F?
- It is a little confusing that the authors choose orange to indicate high potency in Figure 1B and reduced potency in Figure 1F.
- Remove mention of "high-resolution" in line 368 as it is misleading and not necessary.
- In line 341, the authors refer to MERS-CoV and HCoV-NL63 in panel 5C, but this is not correct. I think the MERS-CoV and HCoV-229E panels need switching for this to make sense.
- There appear to be several errors in the reference DOIs.
- In the cryo-EM sample preparation and data acquisition methods, small adjustments could be made to make the methods a

more accurate. Firstly, if 1.19 Å and 0.932 Å pixel sizes were calibrated using a crystal structure, replace “physical pixel size” with “calibrated pixel size.” If these pixel sizes were read from EPU, then you should simply say “a nominal magnification of 105,000x with a pixel size of 1.19 Å.” Physical is often used to refer to the physical pixel size on a detector, which is 14 µm. Secondly, I am assuming that the EER movie was fractionated into a number of dose fractions resulting in ~1 e/Å<sup>2</sup>? If so, it should read “EER movie stack consisting of 1,737 EER frames was dose-fractionated to 50 fractions” or something similar.

- Did the authors really collect the exact same number of movies for the WT and XBB.1.5 data shown in Figure 2A? I can imagine this was a conscious choice to ensure equal sample sizes or simply a typo.
- A general comment is that the writing could be improved. A non-exhaustive list of issues is included below:

Line 24: “this pathogen” implies a single agent.  
Line 55: “of noting” should be “of note.”  
Line 73: “variants has” should be “variants have.”  
Line 97: You can remove the “of” after “sterically hindering.”  
Line 116: “FACS” is written as if it is an abbreviation for flow cytometry.  
Line 662: “Captured” can be replaced with “collected.”  
Extended data line 98: Erroneous punctuation after “and.”

(Remarks on code availability)

Cryo-EM maps and models have been deposited the EMDB and PDB, but I am recommending that raw cryoEM data for two of the data sets should be deposited to EMPIAR.

Version 1:

Reviewer comments:

Referee #1

(Remarks to the Author)

The authors examine a cryptic, conserved epitope that becomes exposed during the conformational changes of fusion.

The most exciting part of the paper was indeed, the fusion intermediate structure. An apo form of this structure was published by some of the authors in Cell. What is different in this paper from the previous is that the intermediate structure is bound by scFv of the antibody of interest. There is also analysis of how Omicron evades such antibodies by some conformational masking (as opposed to mutating amino acid contacts), and by a relative change in protease preference which undoes the ability of the antibody to neutralize by blocking the previously preferred protease.

Lessons applicable to other fields include revelation of antibody recognition of an epitope exposed in a fusion intermediate, exploration of mechanism of antibody escape that are more complex than change in amino acid contact, the finding that smaller versions of the antibody can access cryptic epitopes, and highlighting this particular helix as a potential small molecule target.

Although the aspect I was most excited about - the pre-fusion. intermediate structure, does seem to have been revealed before in an unbound way, the antibody complex and analysis of how antibody is evaded are useful and thoughtful contributions to the field.

I am focusing here on the big picture and not on data specifics.

(Remarks on code availability)

Referee #2

(Remarks to the Author)

I appreciate their extensive effort in addressing the prior comments and improving the quality and clarity of the study. Overall, the authors have appropriately addressed all of my previous concerns and substantially strengthened the manuscript. The inclusion of new structural data, expanded analyses of antibody–spike interactions, and clarification of the “undefined class” add rigor and transparency to the work. Regarding the issue of novelty, I do note some partial overlap with the group’s previously published study. However, this revised manuscript appears to include additional and higher-resolution structural insights, particularly concerning the S2'-helix conformation and mechanisms of antibody evasion by Omicron subvariants. In my view, the revised manuscript is much improved and now provides valuable mechanistic insights that will interest the virology and structural biology communities.

(Remarks on code availability)

Appropriate

Referee #3

(Remarks to the Author)

This reviewer notes that the authors worked diligently to address the extensive reviewers' comments.

This reviewer also provided additional comments below to help the authors strengthen the manuscript, which reports many important findings. Although the relevance of CTSL-mediated entry remains uncertain—given that TMPRSS2-mediated entry appears to be the predominant pathway based on in vivo, ex vivo, and in vitro data—it is nevertheless a thought-provoking study that will be of broad interest.

1. A growing body of in vitro and in vivo data indicate that Omicron variants still use TMPRSS2 and suggest that CTSL utilization is an artifact of some assays (e.g. PMID: 36243815, <https://www.biorxiv.org/content/10.1101/2025.07.07.663433v1>, PMID: 37769654, PMID: 37555660)
2. The role of H655Y as being responsible for the \*apparent\* switch in entry route is debated with other candidate mutations also having supporting data.
3. How confident can the authors be regarding "S2'-helix itself is displaced downward by ~1.2 Å" with a structure determined at worse than 4Å resolution?
4. This reviewer suggests using and showing all biological replicates and carrying out statistical analysis for data shown in Fig1b-c,f and 4c-d
5. The comparison in Fig3 between soluble CTSL and membrane-anchored TMPRSS2 for 76E1 cleavage inhibition may be affected because TMPRSS2 is in cis instead of trans
6. Extended data fig8b should show fit to the data
7. This reviewer agrees that a hydrogen bond is formed between S1 Y655 and S2 T696 in XBB.1.5. However, the hydrogen bond formed between the H655 side chain and A694 in WT S (which the authors presumably referred to, although it requires flipping the H655 side chain 180° in the deposited PDB 6VYB structure) is lost in XBB.1.5
8. Which cell line was used for the HCoV-229E pseudo-virus assay shown in Fig5? Also, it has been established that Vero cells (is it E6, CCL81, another one?) is not a relevant cell line given that it does not faithfully reflect the entry pathway of (at least several strains of) SARS-CoV-2 and leads to loss of the S1/S2 furin cleavage site when used to passage SARS-CoV-2
9. Comparable binding of 76E1 scFv to SARS-CoV and HCoV-229E with vs without receptor in Fig5d is unexpected (the prefusion structure of these spikes also mask the S2' helix).
10. Lines 390-398 should be removed altogether because the cryoEM data are of insufficient quality to speculate on what that undefined class is

(Remarks on code availability)

Referee #4

(Remarks to the Author)

The authors have undertaken significant efforts to address my concerns, as well as those raised by the other reviewers. I am satisfied with the revisions and believe they satisfactorily resolve the issues I previously identified.

(Remarks on code availability)

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# Response to Referees

Dear Reviewers:

We sincerely thank you very much for your thorough evaluation of our manuscript and your constructive comments. In this revised version, we have carefully addressed all comments, either by incorporating new data that further supports our original conclusions or by providing a clear rationale when certain requests went beyond the scope of this study.

We recognize that several reviewers raised similar concerns regarding the structural section of our manuscript. In response, we have substantially revised the structure section and added extensive new experimental data to address these points:

## **(1) Structural comparison of the 76E1-bound WT S<sup>E-FIC</sup> and apo D614G S<sup>E-FIC</sup>.**

We compared the 76E1-bound WT S<sup>E-FIC</sup> with the previously reported apo D614G S<sup>E-FIC</sup> <sup>1</sup> (Cell, 2025, PMID: 39889696) and found that 76E1 specifically recognizes an open conformation of S<sup>E-FIC</sup>. There is not enough room for 76E1 binding in apo E-FIC. Substantial conformational rearrangements are required to expose the 76E1 epitope, including a downward movement and counterclockwise rotation of the S1-ACE2 along the threefold axis, followed by outward rotation of ACE2 (**Extended Data Fig. 6g-k and Extended Data Fig. 7a**, see below.). Notably, the 76E1 epitope adopts a helical conformation that we designated the S2'-helix, which corresponds to a loop in the apo E-FIC structure.

## **(2) We determined the cryo-EM structure of XBB.1.5 S<sup>E-FIC</sup>-76E1.**

To overcome the orientation preference that precluded structure determination of the XBB.1.5 S<sup>E-FIC</sup>-76E1 complex in the previous dataset, we optimized the cryo-EM sample preparation. Specifically, we prepared two grids of the complex, with or without 2% glycerol, and collected new cryo-EM datasets. The datasets were combined for 3D reconstruction, yielding an overall structure at 5.65 Å resolution with C3 symmetry. Subsequent local refinement of the S2-76E1 and top S2-76E1 regions produced

reconstructions at 4.14 Å resolution, enabling structural interpretation and analysis  
(Extended Data Fig. 5).

Structural comparison between XBB.1.5 S<sup>E-FIC</sup>-76E1 and WT S<sup>E-FIC</sup>-76E1 reveals  
that XBB.1.5 S partially evade 76E1 binding through conformational shifts and  
rotations of the S2'-helix, and accompanied by a subtle upward displacement of the S1-  
ACE2 modules in the E-FIC. These changes introduce steric hindrance at the 76E1-  
binding interface, necessitating a more pronounced outward expansion of ACE2s to  
sufficiently expose the S2'-helix, which likely imposes a higher energetic barrier to  
76E1 engagement (Fig. 2 and Extended Data Fig. 7a).

**(3) We tested the impact of 76E1 on the structural transition of WT, D614G, and  
XBB.1.5 S following ACE2 priming.**

WT, D614G, and XBB.1.5 S were mixed with ACE2 and 76E1-Fab or scFv at molar  
ratios of 1:1.2:1.2, respectively and incubated on ice for 0 min, 10 min, 30 min, and 120  
min. Structural transitions were assessed by negative-staining electron microscopy at  
each time point. Samples of S alone and S-ACE2 (molar ratio: 1:1.2) were prepared as  
control.

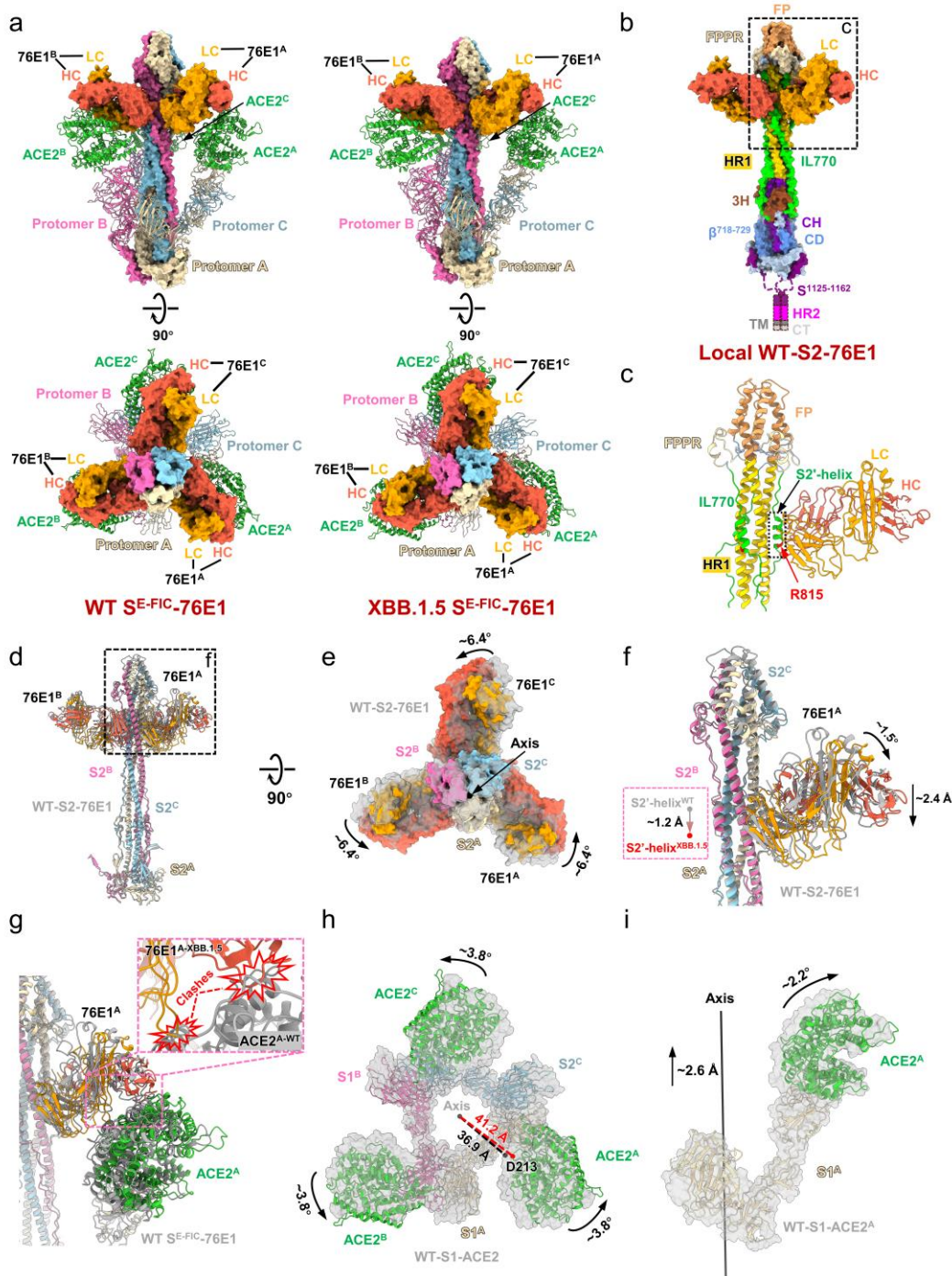
Compared to WT and D614G S, XBB.1.5 S exhibited reduced conformational  
stability. Upon ACE2 binding, all S transitioned to the postfusion state. In contrast,  
postfusion particles were rarely observed in samples containing either 76E1-Fab or  
scFv (Supplementary Fig. 1-3).

[REDACTED]

[REDACTED]

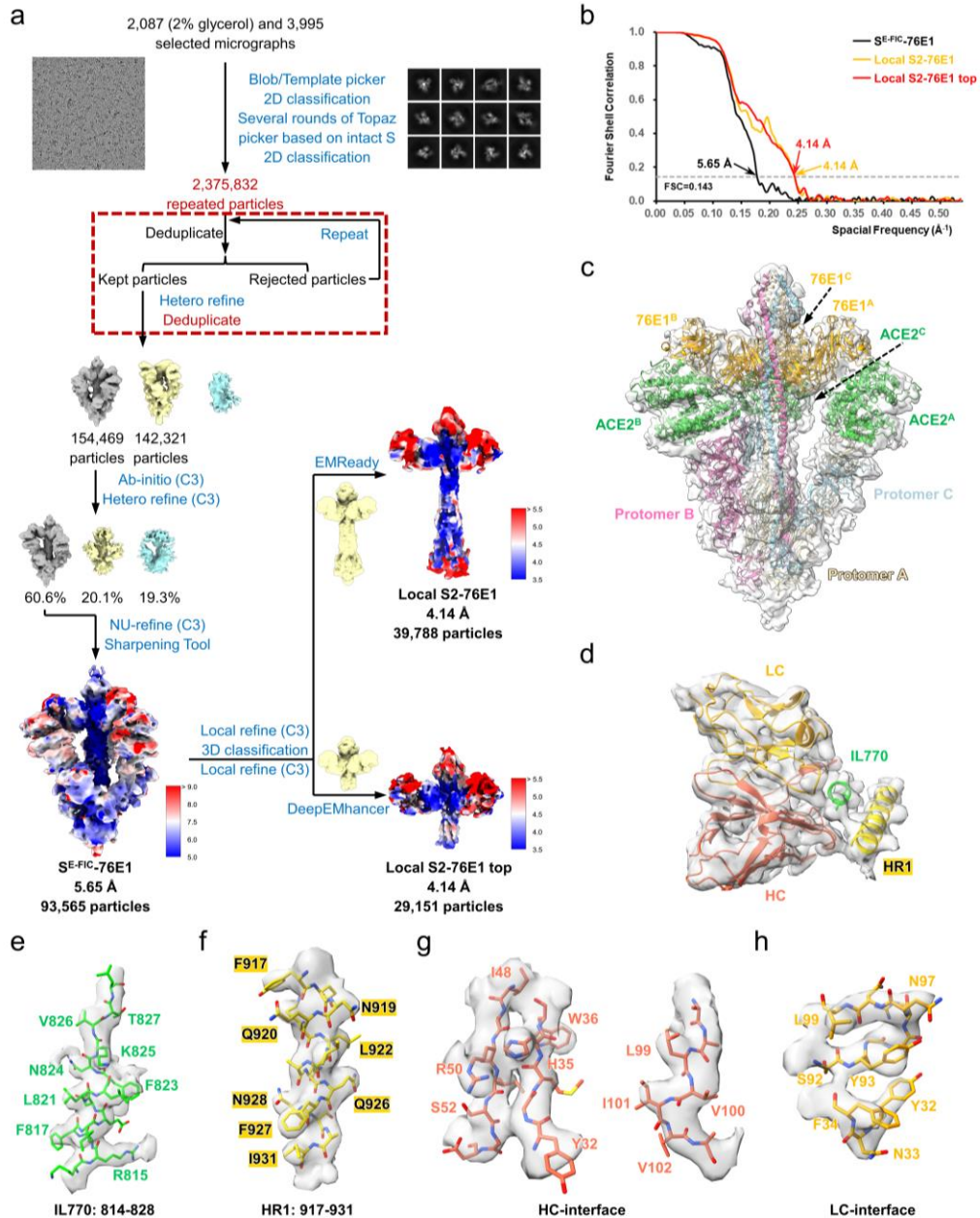
Given its minimal contribution to our main findings, we have removed detailed descriptions of this class from the *Main text* and instead address it briefly in the *Discussion* section of the revised manuscript. We will further clarify the post-S1-ACE2 shedding conformational transitions of the spike, with or without 76E1, in future work. We appreciate the reviewers' understanding of these limitations.

Below, we present the reorganized figures included in the revised manuscript (Given the extensive structural data and the overlapping concerns raised by multiple reviewers, we have consolidated all structural figures into an integrated presentation below):



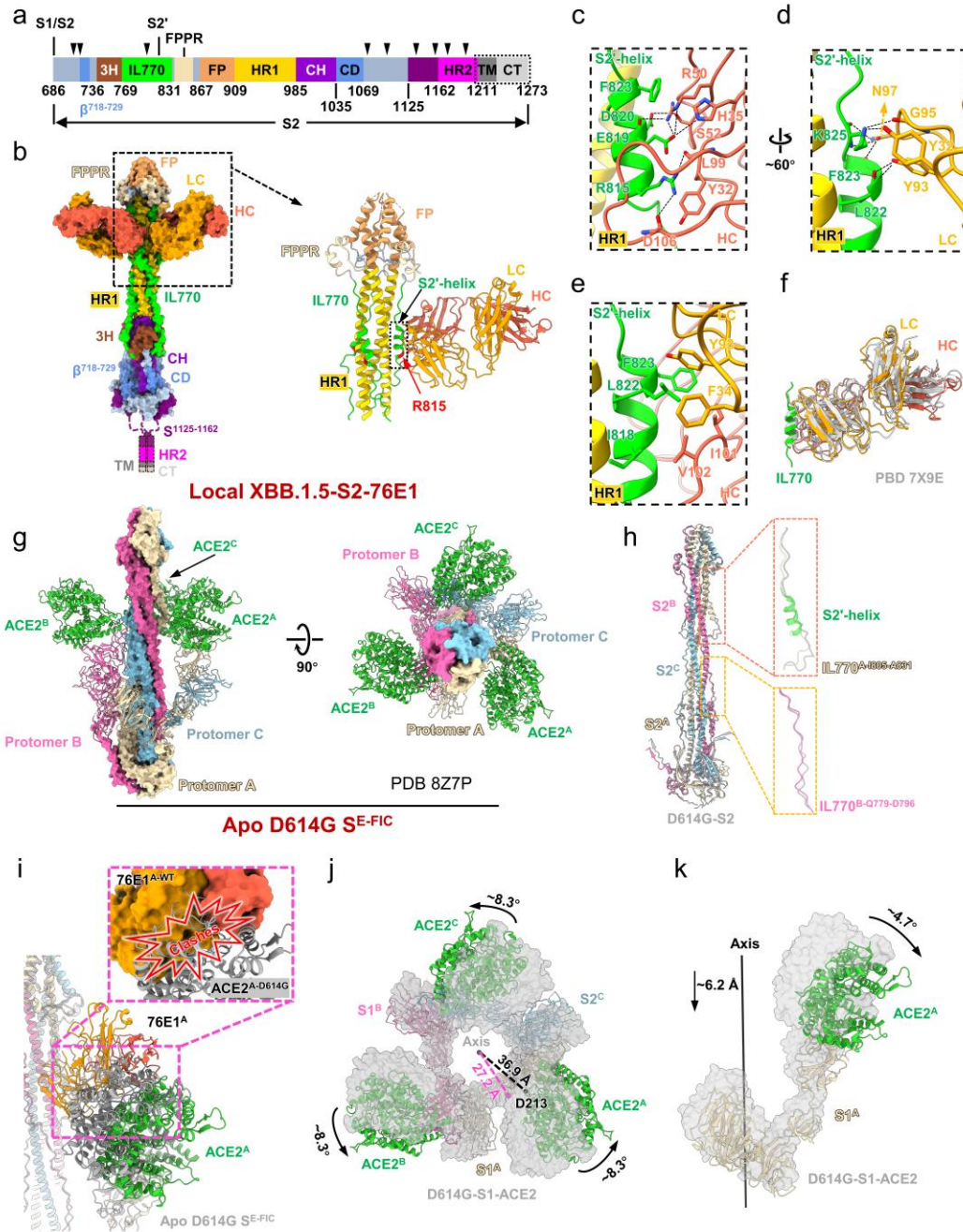
**Fig.2 Molecular mechanisms of partial evasion of Omicron variants from 76E1.** **a**, The overall structures of the WT S<sup>E-FIC</sup>-76E1 (left) and XBB.1.5 S<sup>E-FIC</sup>-76E1 (right) with S1-ACE2 and S2-76E1-Fab shown as ribbon and surface, respectively. Various components are colored in wheat (protomer A), hot pink (protomer B), sky blue (protomer C), and lime green (ACE2 molecules) with heavy chain (HC) and light chain (LC) of 76E1-Fab shown in tomato and orange, respectively. **b**, Surface representation of the local WT-S2-76E1 structure with S2 colored as in **Extended Data Fig. 6a** and 76E1 in **(a)**. **c**, Magnified view of the dashed boxed region in **(b)**. The S2'-helix epitope is

indicated by a dashed line and key residue R815 is in red. **d**, Structural comparison of S2-76E1 in WT S<sup>E-FIC</sup>-76E1 and XBB.1.5 S<sup>E-FIC</sup>-76E1. **e**, Same as (**d**), but with a 90° rotation. Arrows in arcs show the rotations of the S2 top region and 76E1 from WT S<sup>E-FIC</sup>-76E1 to XBB.1.5 S<sup>E-FIC</sup>-76E1. **f**, Magnified view of the dashed boxed region in (**d**). Arrows in arcs show 76E1 rotating outward ~1.5° from the axis relative to CH and CD centroid, with arrows indicating the movements of S2'-helix or 76E1. **g**, Collisions occur between the 76E1 in XBB.1.5 S<sup>E-FIC</sup>-76E1 and the ACE2 in WT S<sup>E-FIC</sup>-76E1. **h**, Structural comparison of trimeric S1-ACE2 in WT S<sup>E-FIC</sup>-76E1 (surface) and XBB.1.5 S<sup>E-FIC</sup>-76E1 (ribbon). Arrows show the rotations of S1-ACE2s, and dashed lines indicate minimum distances between the S 3-fold axis and ACE2 (D213). **i**, Same as (**h**), but only S1-ACE2<sup>A</sup> is shown with a 90° rotation; meanwhile, S1-ACE2<sup>A</sup> in WT S<sup>E-FIC</sup>-76E1 rotates ~3.8° counterclockwise around the 3-fold axis. Arrows in arcs show the ACE2's rotations relative to the trimeric SD2 centroid, after the S1-ACE2' rotating counterclockwise (~3.8°) and shifting upward (~2.6 Å). Arrows indicate the movements of S1-ACE2. In (**d-i**), the WT S<sup>E-FIC</sup>-76E1 is in gray, and the XBB.1.5 S<sup>E-FIC</sup>-76E1 is colored as in (**a**). These structures are aligned by invariant CD and CH.



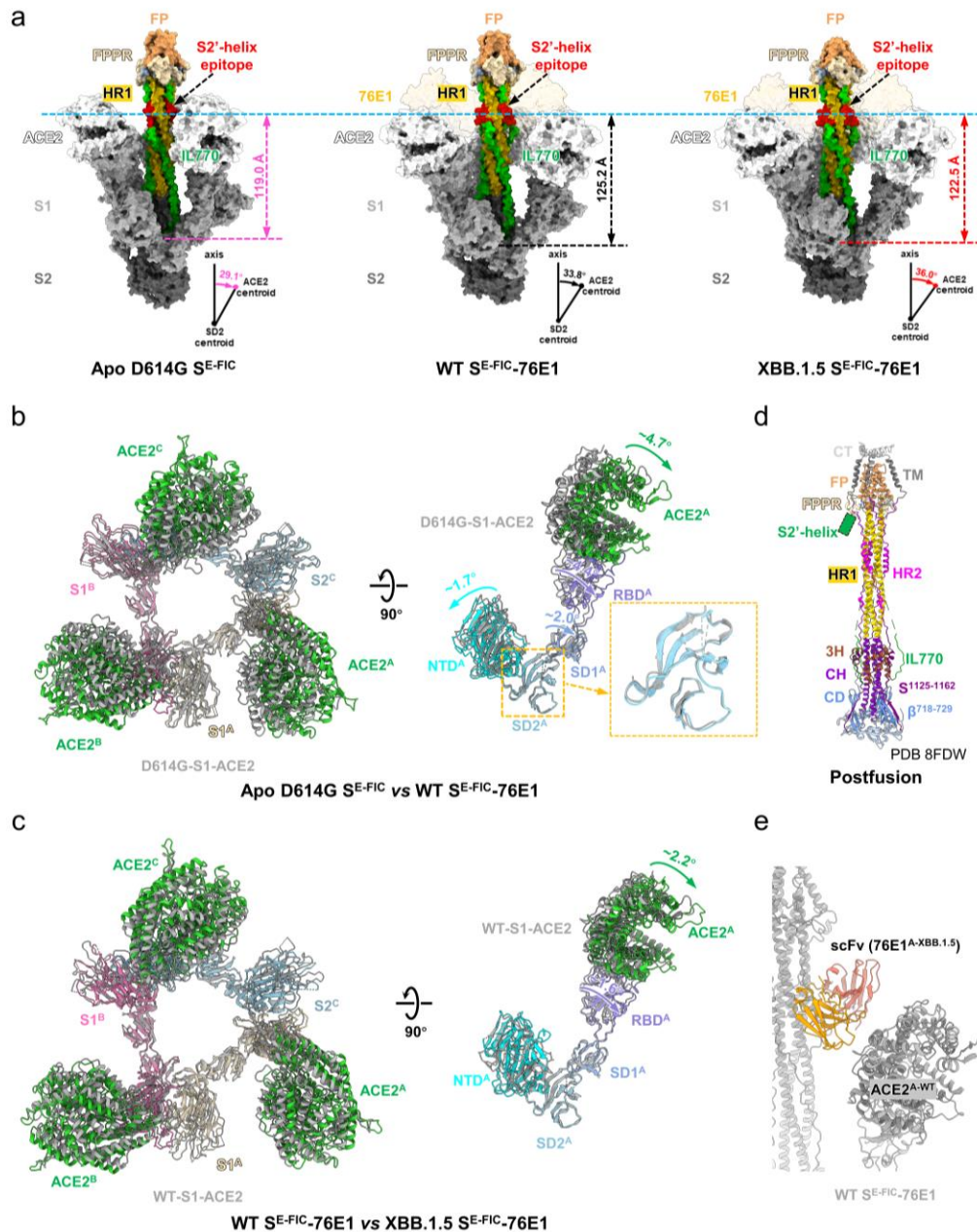
**Extended Data Fig. 5 Cryo-EM workflow and map quality analysis of XBB.1.5-ACE2-76E1**

**Fab complex.** **a**, Cryo-EM data processing workflow. The final cryo-EM maps are colored according to the local resolution estimated by cryoSPARC with the FSC=0.143 criterion. **b**, Gold standard FSC curves of the refined 3D reconstructions in XBB.1.5 S<sup>E-FIC</sup>-76E1. **c**, The XBB.1.5 S<sup>E-FIC</sup>-76E1 structure fits into the 5.65 Å map. **d**, The S2-helix in IL770 and variable regions of 76E1-Fab fit into the local S2-76E1 map at a resolution of 4.14 Å. **e-h**, Representative density in gray surface representation from the local S2-76E1 map. Colors of various components in **c-h** correspond to those in **Extended Data Fig. 4d-i**, respectively.



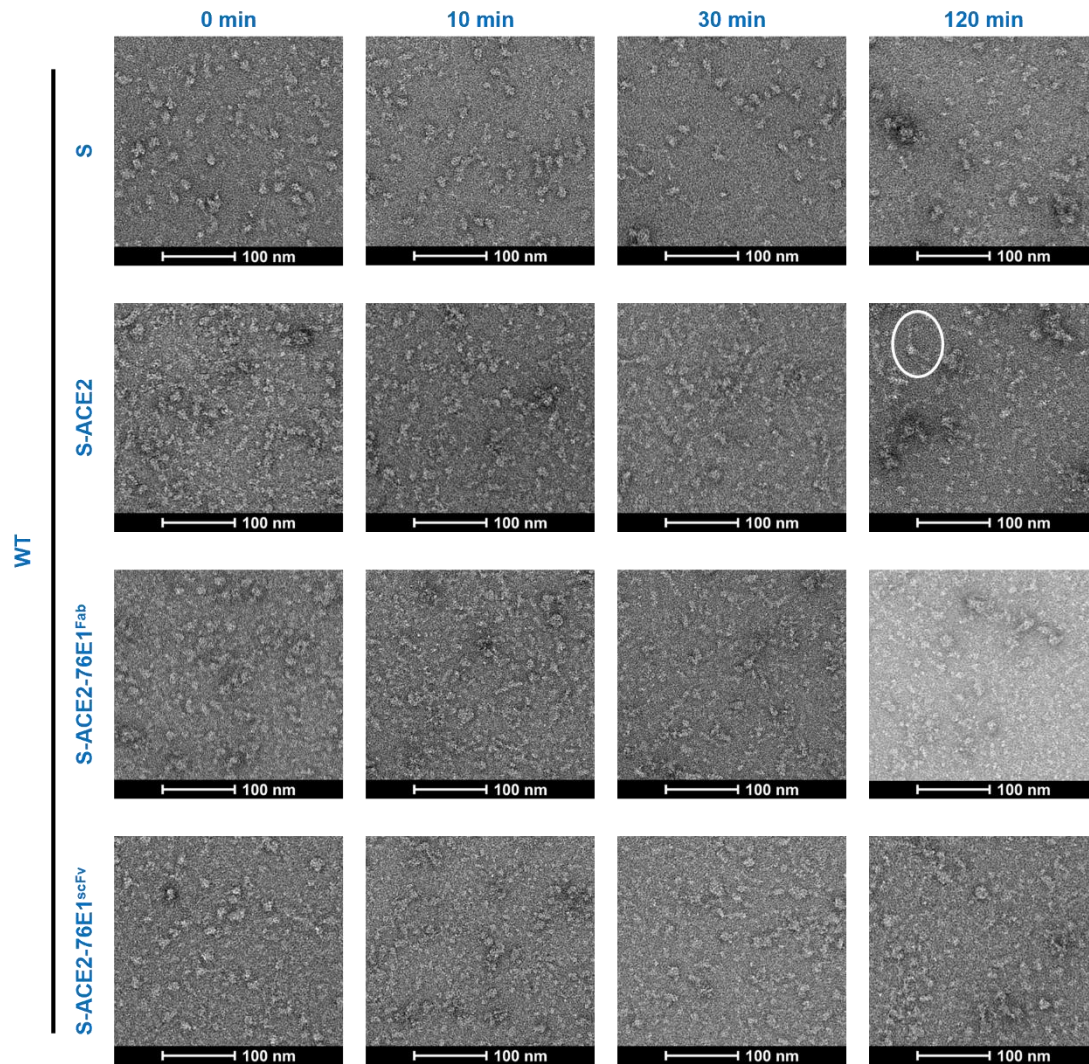
**Extended Data Fig. 6 Neutralization mechanism of 76E1 revealed by comparing the WT S<sup>E</sup>-FIC-76E1 structure with apo D614G S<sup>E</sup>-FIC.** **a**, Schematic of the SARS-CoV-2 S2 colored by domain. Segments that were excluded from the ectodomain expression construct are indicated with dashed boxes. Segments include S1/S2, S2' cleavage site; β<sup>718-729</sup>; 3H, three-helix segment; IL770, intermediate loop 770; FPPR, fusion peptide proximal region; FP, fusion peptide; HR1, heptad repeat 1; CH, central helix; CD, connector domain; HR2, heptad repeat 2; TM, transmembrane anchor; CT, cytoplasmic tail; and inverted triangles for glycans in S2. **b**, Surface representation of the local XBB.1.5-S2-76E1 structure (left) and magnified view of the dashed boxed region (right),

114 with S2 colored as in (a) and 76E1 in **Fig. 2a**. The S2'-helix epitope is indicated by a dashed line  
 115 and key residue R815 is in red. **c-e**, Interfaces between 76E1-Fab and the S2'-helix in the WT S<sup>E-FIC</sup>-  
 116 76E1 structure. Hydrogen bonds between S2'-helix and HC (**c**) or LC (**d**) of 76E1-Fab, and  
 117 hydrophobic interaction (**e**) between S2'-helix and 76E1-Fab are showed. Key residues are shown  
 118 as a stick model and hydrogen bonds are indicated by dotted lines. **f**, Structural comparison of the  
 119 WT S2'-helix-76E1 structure resolved by cryo-EM (colored as in (b)) and crystallography (PDB  
 120 7X9E, gray). **g**, The overall structure of the apo D614G S<sup>E-FIC</sup> (PDB 8Z7P) showed as in **Fig. 2a**. **h**,  
 121 Left: Structural comparison of S2 in apo D614G S<sup>E-FIC</sup> (gray) and WT S<sup>E-FIC</sup>-76E1 (colored as in  
 122 **Fig. 2a**). Right: The magnified view of the dashed boxed region indicates a slight movement of the  
 123 flexible area in IL770. **i**, Collisions occur between the 76E1 in WT S<sup>E-FIC</sup>-76E1 and the ACE2 in  
 124 apo D614G S<sup>E-FIC</sup>. **j**, Structural comparison of trimeric S1-ACE2 in apo D614G S<sup>E-FIC</sup> (surface) and  
 125 WT S<sup>E-FIC</sup>-76E1 (ribbon). Arrows show the rotations of S1-ACE2s, and dashed lines indicate  
 126 minimum distances between the S 3-fold axis and ACE2 (D213). **k**, Same as (j), but only S1-ACE2<sup>A</sup>  
 127 is shown with a 90° rotation; meanwhile, S1-ACE2<sup>A</sup> in apo D614G S<sup>E-FIC</sup> rotates ~8.3°  
 128 counterclockwise around the 3-fold axis. Arrows in arcs show the ACE2's rotations relative to the  
 129 trimeric SD2 centroid, after the S1-ACE2' rotating counterclockwise (~8.3°) and shifting downward  
 130 (~6.2 Å). Arrows indicate the movements of S1-ACE2. In (h-k), the apo D614G S<sup>E-FIC</sup> is in gray  
 131 color and the WT S<sup>E-FIC</sup>-76E1 is colored as in **Fig. 2a**. These structures are aligned by invariant CD  
 132 and CH.

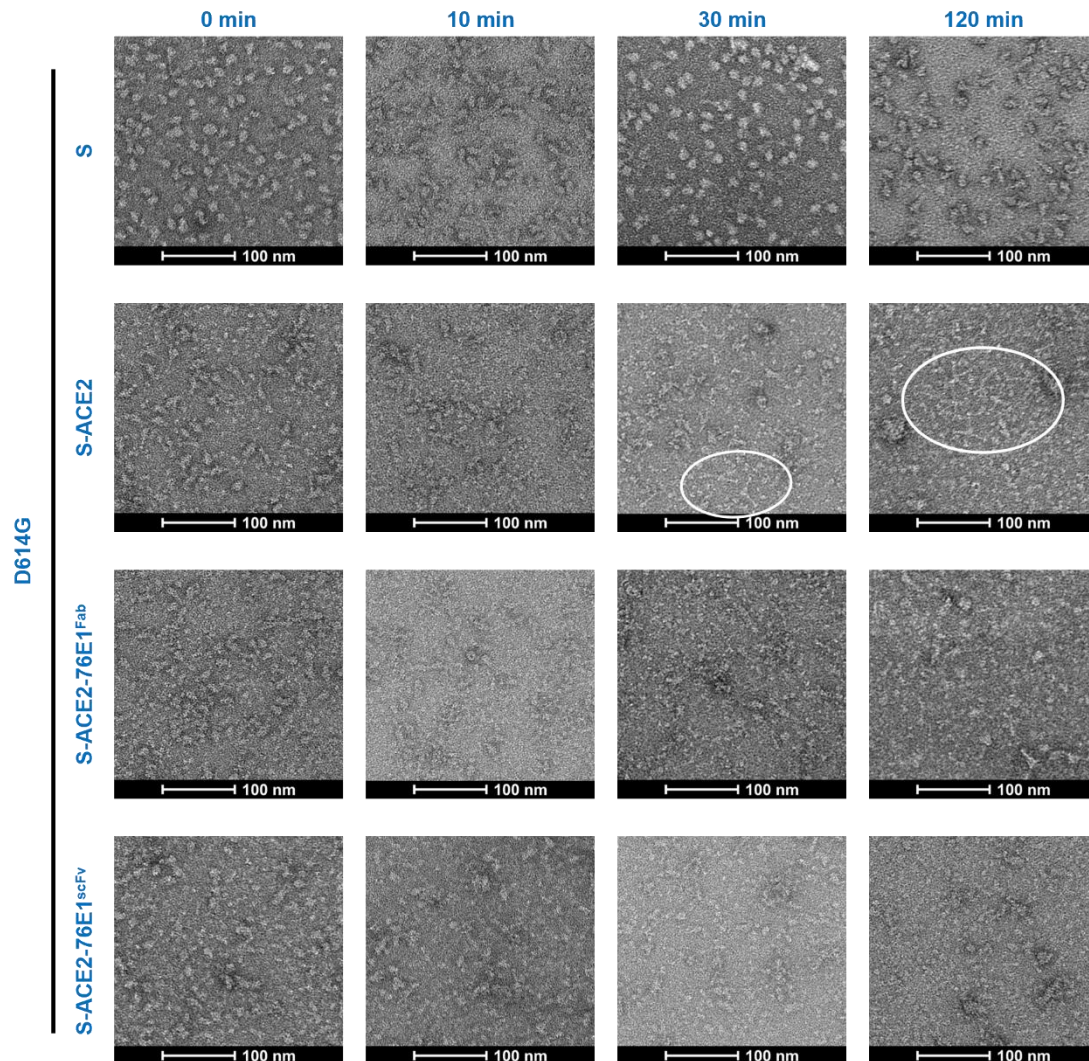


**Extended Data Fig. 7 Additional details of the apo D614G S<sup>E-FIC</sup>, WT S<sup>E-FIC</sup>-76E1 and XBB.1.5 S<sup>E-FIC</sup>-76E1 structures.** **a**, Surface representation of the apo D614G S<sup>E-FIC</sup>, WT S<sup>E-FIC</sup>-76E1 and XBB.1.5 S<sup>E-FIC</sup>-76E1 structures. ACE2, 76E1 and S1 are colored in white, orange, and dark gray, respectively. The S2 is in dim gray, of which the IL770, FPPR, FP and HR1 are colored as in **Extended Data Fig. 6a**, with S2'-helix in red. The distance is defined by the gap between the plane containing the centroid of each S2'-helix and another plane containing the centroid of each SD2. Using the trimeric SD2 centroid as the vertex, the angle is defined by the deflection of one ACE2 molecule centroid in relation to the 3-fold axis. **b**, Left: Top view of the superposition of the trimeric S1-ACE2 structures from apo D614G S<sup>E-FIC</sup> (gray) and WT S<sup>E-FIC</sup>-76E1 (colored as in **Fig 2a**) after

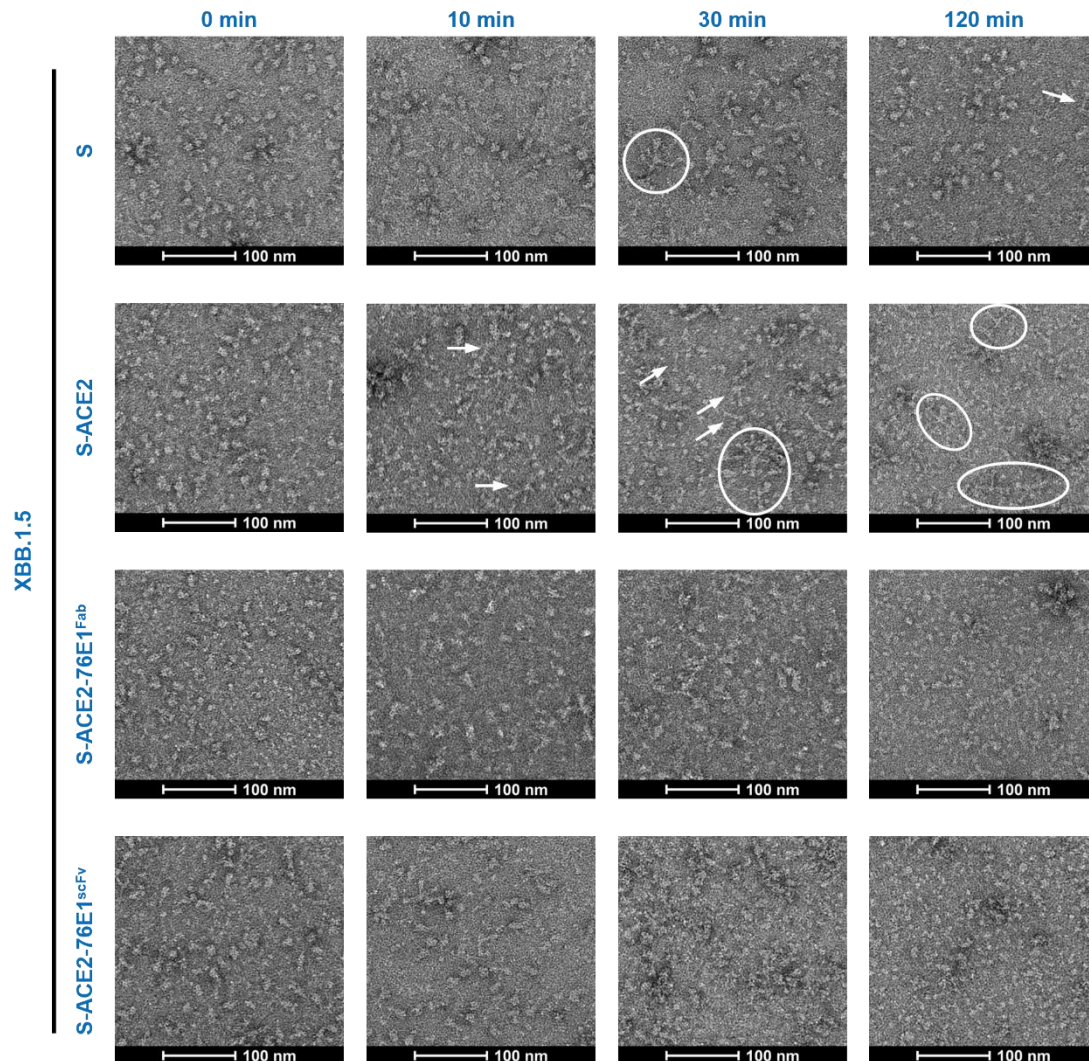
the D614G-S1-ACE2 rotating ( $\sim 8.3^\circ$ ) around the axis and shifting downward ( $\sim 6.2$  Å). Right: Side view of the superposition of the S1-ACE2<sup>A</sup>. Arrows in arcs show the rotations of ACE2<sup>A</sup> and each domain in S1<sup>A</sup>, relative to the trimeric SD2 centroid, while the SD2 domain remains unchanged. **c**, Left: Top view of the superposition of the trimeric S1-ACE2 structures from WT S<sup>E-FIC</sup>-76E1 (gray) and XBB.1.5 S<sup>E-FIC</sup>-76E1 (colored as in **Fig 2a**) after the WT-S1-ACE2 rotating ( $\sim 3.8^\circ$ ) around the axis and shifting upward ( $\sim 2.6$  Å). Right: Side view of the superposition of the S1-ACE2<sup>A</sup>. Arrows in arcs show the rotations of ACE2<sup>A</sup> and each domain in S1<sup>A</sup> relative to the trimeric SD2 centroid. **d**, The overall structure of the S2 trimer in the postfusion conformation (PDB 8FDW) shown as a ribbon diagram, colored as in **Extended Data Fig. 6a**. The S2'-helix epitope is indicated by a dashed box colored in lime. **e**, Clashes between the 76E1-Fab from XBB.1.5 S<sup>E-FIC</sup>-76E1 and the ACE2 from WT S<sup>E-FIC</sup>-76E1 can be avoided by reducing the antibody size into scFv.



**Supplementary Fig. 1 Impact of varying sizes of 76E1 on the WT S transition after ACE2 priming.** The 76E1-Fab and scFv could effectively block the transition of ACE2-primed WT S to the postfusion conformation.



**Supplementary Fig. 2 Impact of varying sizes of 76E1 on the D614G S transition after ACE2 priming.** ACE2-activated D614G S could rapidly form a postfusion conformation, which might be hindered by 76E1-Fab and scFv.



**Supplementary Fig. 3 Impact of varying sizes of 76E1 on the XBB.1.5 S transition after ACE2 priming.** The XBB.1.5 S was less stable and ACE2-activated XBB.1.5 S could rapidly form a postfusion conformation. Few postfusion particles were observed in the mixtures containing 76E1-Fab and scFv, comparison to the S-ACE2 mixture.

[REDACTED]

Moreover, we have conducted additional functional experiments as the reviewers suggested, which further enhanced the quality of this manuscript. For the revised functional experimental figures, we present them after each specific reviewer comment point-by-point.

**(1) We revised the text description to avoid the misunderstanding that Omicron escapes 76E1 by actively evading immunity.** Our intention was to convey that the Omicron variants have undergone changes in its entry pathway, which subsequently resulted in a reduced neutralizing activity of these antibodies, rather than the antibodies driving the alteration in the virus's entry pathway.

**(2) To quantitatively analyze all Western blot (WB) data, dose-dependent WB experiments were performed. Band intensities were quantified to determine:** ① Antibody binding ratios (scFv/Fab): calculated from scFv-his and Fab-his band intensities (**Fig. 1e, 4d, 5d**) to assess enhanced spike-binding activity after antibody size reduction, indicating potential steric hindrance; ② S2' cleavage proportion

(S2'/(S2'+S2)): calculated from S2' band intensities (**Fig. 3a-b**) to measure TMPRSS2- or CTSL-mediated cleavage efficiency at the S2' site; ③ S2' cleavage reduction rates: quantified in 76E1-treated groups (**Fig. 3c-d**) to determine 76E1's inhibitory efficiency on TMPRSS2- or CTSL-mediated S2' cleavage. S1 shedding served as the sample control in all WB experiments.

**(3) The BF.7 pseudovirus with Y655H reversion was included as the control to assess entry pathways and S2'-helix accessibility.** These data further confirm that H655Y mediates 76E1 evasion via epitope shielding and altered entry pathways. This effect can be reversed by using small-sized antibody formats (**Fig. 4c-f**).

**(4) We further broadened our investigation into the S2'-helix accessibility across other human coronaviruses** by comparing scFv/Fab binding to cell-expressed S proteins. Coronavirus-specific steric hindrance was observed: substantial in SARS-CoV-1 and HCoV-229E, but absent in MERS-CoV and HCoV-NL63 (**Fig. 5d**).

**(5) Additional methodological refinements include:** Extended incubation time of antibody and varying pH conditions to determine whether acidic pH significantly influences antibody dissociation (**Extended Data Fig. 8c-d**); Normalization of cell-surface S protein expression (WT vs. Omicron) using the S3H3 antibody, which exhibits no escape against Omicron variants (**Fig. 1c**); Utilization of the BF.7 strain rather than BA.1 for acid-dependent ELISA and BLI assays (**Extended Data Fig. 8a-b**).

To better highlight the innovations of this manuscript, the title has been modified to “*Conformational change of Omicron spike in an opened early fusion intermediate sterically impairs pan-coronavirus antibody access*”.

The corrections have been highlighted in yellow in the revised manuscript. Following, we address the comments of the four referees in detail.

219 **Referee #1 (Remarks to the Author):**

220 This work studies antibodies against the S2' helix, an area of interest as it is  
221 conserved across coronaviruses. This epitope is exposed after ACE2 priming. An  
222 interesting question is why does the neutralization capacity of these antibodies of  
223 interest diminish in Omicron given that Omicron contains no mutations in the  
224 epitope? Another remaining question is how antibodies against this epitope  
225 achieve their neutralization – presumably blocking of conformational change - but  
226 exactly how is unknown. At the end of the day, these authors find through binding  
227 studies and cryoEM, that Omicron mutations elsewhere decrease access to this  
228 epitope, and that Omicron further evades these antibodies by switching to a  
229 cathepsin-mediated entry pathway.

230 1, Notable structural biology studies include elucidation of intermediates in the  
231 refolding pathway of spike- I see the early intermediate structures, but not the late ones.  
232 Did I miss them in the paper?

233 **Response:**

234 We apologize for the inconvenience. We previously named a low-resolution  
235 reconstruction as late intermediate state. However, after careful analysis, we renamed  
236 this class as “undefined class” because it may present a late state or a broken early state  
237 (Supplemental Fig. 4). We address it briefly in the *Discussion* section of the revised  
238 manuscript. (Lines 391-399)

239 Some other observations are that Omicron seems to refold faster than WT,  
240 and that 76E1 can't completely inhibit refolding to postfusion in Omicron,  
241 although it could in WT.

242 Their findings suggest that scFv will more effectively target this site than full  
243 antibodies, and that accessibility of the site to the proteases might also influence  
244 infectivity.

245 One of the Omicron mutations H655Y reduces neut by these antibodies – the

authors find that it must limit epitope exposure as scFv neutralizes H655Y-bearing spike more than Fab. They analyze contacts in previous structures and suggest that this mutation may alter the size of the shift or propensity of S1 to shift from S2 which alters epitope exposure. Comparison to crystal structures suggests that the orientation of the different antibodies influences their ability to bind and neutralize. Further, Omicron uses cathepsin mediated entry in the endosome.

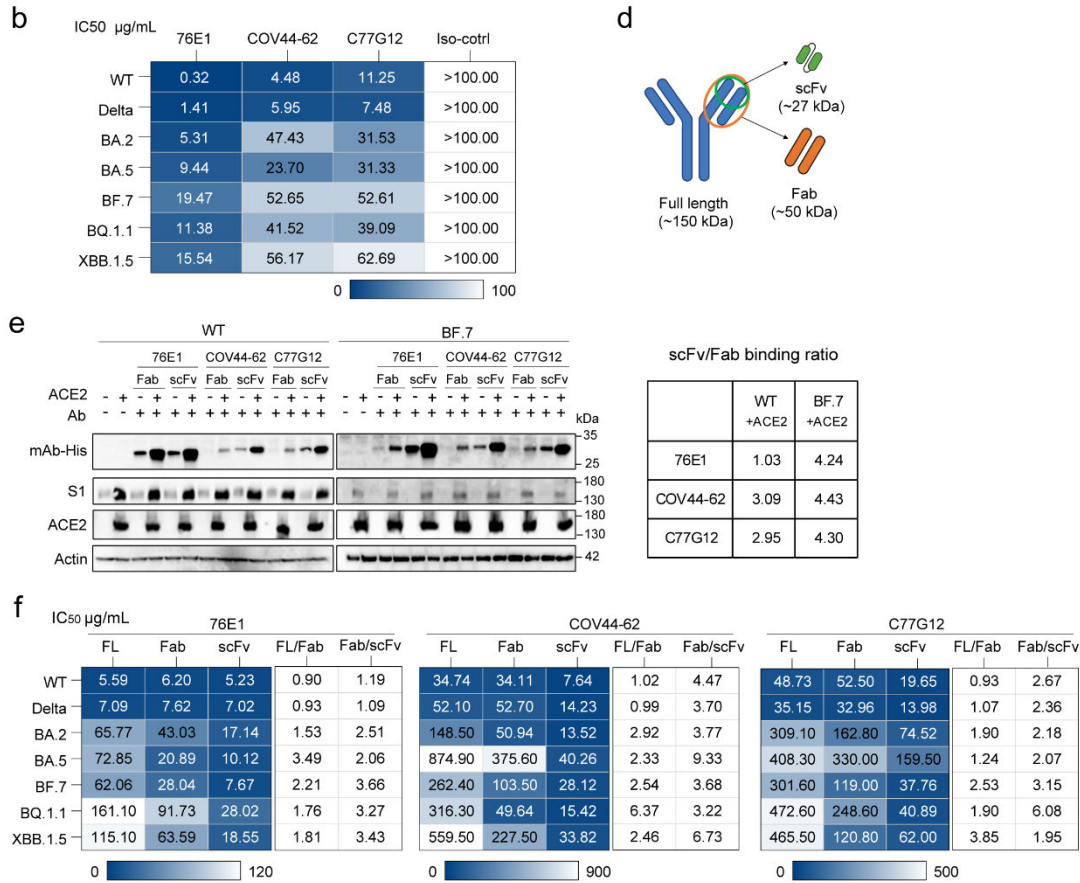
2, I'm not fully understanding the experiments of 1b and 1d, but it is clear that scFv neutralizes better than Fab which does suggest an epitope access issue. scFv neuts better than Fab against other coronaviruses as well.

**Response:**

We sincerely appreciate your valuable feedback regarding this figure. To better understand this figure for the reviewers and readers, we have carefully modified the data presentation and description in the revised manuscript. Specially, the scFv/Fab binding ratios, and FL/Fab and Fab/scFv IC50 ratios were additionally included to enable intuitive assessment of binding and neutralization activity across antibody formats of varying sizes.

**Fig. 1b** now clearly demonstrates that three S2'-helix-targeting bnAbs (76E1, COV44-62, C77G12) exhibit reduced neutralization against Omicron (elevated IC50 values) despite epitope conservation. To investigate this evasion mechanism, we conducted experiments in **Fig. 1d-g**. This cryptic epitope buried in the prefusion spike and becomes accessible upon ACE2 engagement, led us hypothesized enhanced steric occlusion at S2' sites in Omicron S than the WT S. To validate this hypothesis, we constructed two his-tagged antibody formats varying in molecular size: the Fab fragment (~50 kDa) and scFv fragment (~27 kDa) (**Fig. 1d and Extended Data Fig. 1a**). 76E1-Fab and scFv fragments demonstrated comparable binding ability towards the WT S, with an scFv/Fab binding ratio of 1.03. In contrast, 76E1-scFv exhibited enhanced binding activity to Omicron spike than Fab, evidenced by an increased scFv/Fab binding ratio of 4.24. These findings indicate that the Omicron spike is likely

274 have more steric hindrance to the large-sized 76E1-Fab, whereas this spatial constraint is  
275 absent in the WT S. In the case of COV44-62 and C77G12, the scFv/Fab binding ratio  
276 to Omicron S is slightly higher than that of WT spike (**Fig. 1e**). These observations  
277 were further validated in a pseudovirus neutralization assay using 293T-ACE2 cells.  
278 The WT S2' epitope showed similar accessibility for 76E1-FL, Fab and scFv against  
279 WT and Delta viruses, indicated by comparable IC<sub>50</sub> ratios. However, significant steric  
280 hindrance was observed for the Omicron S2' epitope with the 76E1-FL and Fab, as the  
281 neutralizing activity increased markedly when reducing to Fab and scFv, reflected by  
282 elevated FL/Fab and Fab/scFv IC<sub>50</sub> ratios, as compared with the WT. For COV44-62  
283 and C77G12, reduction from FL to Fab minimally affected WT pseudovirus  
284 neutralization but significantly enhanced Omicron neutralization, indicating enhanced  
285 steric hindrance at the Omicron S2' site to FL of COV44-62 and C77G12. Further  
286 reduction to scFv markedly increased the neutralizing activity against both WT and  
287 Omicron (**Fig. 1f and Extended Data Fig. 2**). Collectively, these data conclusively  
288 demonstrate that antibody size directly modulates their accessibility to this epitope.  
289 Omicron displays greater steric occlusion at its S2' site than WT, which directly affects  
290 antibody binding to this epitope. (**Lines 110-149**).



**Fig.1 Omicron demonstrated enhanced steric hindrance to antibodies targeting the spike<sup>815-825</sup>.** **b**, Neutralizing IC<sub>50</sub> values of full-length 76E1, COV44-62 and C77G12 against SARS-CoV-2 WT, Delta and Omicron variant spike protein-pseudotyped viruses, as evaluated in 293T-ACE2 cells. **d**, Schematic graphic of antibody formats with varying molecular size: 76E1-FL, Fab and scFv. **e**, Western blot analysis of scFv and Fab binding activity at equal molar concentrations to cell-surface-expressed WT and BF.7 S proteins, with or without ACE2-Fc. His-tagged antibody formats were detected using anti-His-HRP antibody. S1 subunit shedding into the cell culture supernatant was quantified. ScFv/Fab binding ratios under ACE2 pre-incubation were calculated through quantitative assessment of band intensities. **f**, Neutralizing IC<sub>50</sub> values of 76E1, COV44-62, and C77G12 (equal molar concentrations; varying molecular sizes) against SARS-CoV-2 WT, Delta, and Omicron pseudoviruses in 293T-ACE2 cells. (**b**, **f**) For each hCoV pseudovirus assay, all antibodies were compared in parallel. For all data, one representative experiment of at least two is shown.

**3**, I have focused on big picture here rather than experimental detail. Any exploration of the spike epitope conserved across coronaviruses and how to make recognition of

these epitopes more successful is good. The finding that they are compromised in access and some insight into why will also guide some eventual therapeutic utility. Of chief importance is that fusion intermediate structure, although I wish I had seen a structure of the later intermediate, even a low resolution one, as well. I see a light blue trimer in extended data figure 3 - but this density may be three copies of the antibody and not the spike intermediate. These structures are difficult to achieve but would be notable if possible.

**Response:**

We appreciate the reviewer's emphasis on the broader significance of the work, which is indeed a key focus of our study.

Response to “Any exploration of the spike epitope conserved across coronaviruses and how to make recognition of these epitopes more successful is good”: we have added extensive new experimental data to address this point.

**(1) Structural comparison of the 76E1-bound WT S<sup>E-FIC</sup> and apo D614G S<sup>E-FIC</sup>.**

We compared the 76E1-bound WT S<sup>E-FIC</sup> with the previously reported apo D614G S<sup>E-FIC</sup> <sup>1</sup> (Cell, 2025, PMID: 39889696) and found that 76E1 specifically recognizes an open conformation of S E-FIC. There is not enough room for 76E1 binding in apo E-FIC. Substantial conformational rearrangements are required to expose the 76E1 epitope, including a downward movement and counterclockwise rotation of the S1-ACE2 along the threefold axis, followed by outward rotation of ACE2 (**Extended Data Fig. 6g-k and Extended Data Fig. 7a**). Notably, the 76E1 epitope adopts a helical conformation that we designated the S2'-helix, that corresponds to a loop in the apo E-FIC structure.

**(2) We determined the cryo-EM structure of XBB.1.5 S<sup>E-FIC</sup>-76E1.**

We resolved the overall structure of XBB.1.5 S<sup>E-FIC</sup>-76E1 to 5.65 Å resolution with C3 symmetry. Local refinement of the S2-76E1 and top S2-76E1 regions yielded reconstructions at 4.14 Å resolution, enabling detailed structural analysis. (**Extended**

**Data Fig. 5).** Structural comparison between XBB.1.5 S<sup>E-FIC</sup>-76E1 and WT S<sup>E-FIC</sup>-76E1 reveals that XBB.1.5 S partially evade 76E1 binding through conformational shifts and rotations of the S2'-helix, and accompanied by a subtle upward displacement of the S1-ACE2 modules in the E-FIC. These changes introduce steric hindrance at the 76E1-binding interface, necessitating a more pronounced outward expansion of ACE2s to sufficiently expose the S2'-helix, which likely imposes a higher energetic barrier to 76E1 engagement (**Fig. 2 and Extended Data Fig. 7a**).

**(3)** We further broadened our investigation into the assessment of the S2'-helix accessibility across other human coronaviruses by comparing scFv/Fab binding to cell-expressed spike proteins. Coronavirus-specific steric hindrance was observed: substantial in SARS-CoV-1 and HCoV-229E, but absent in MERS-CoV and HCoV-NL63 (**Fig. 4c-f**).

**(4)** We further conducted dose-dependent experiments of TMPRSS2/CTSL cleavage and antibody inhibition with IC<sub>50</sub> calculations for WT and BF.7. Omicron reduced efficiency of TMPRSS2-mediated S2' cleavage, which may explain Omicron's altered entry pathway preference. The increased utilization of CTSL-dependent entry pathways of Omicron contributes to immune evasion to 76E1.

[REDACTED]

[REDACTED]

## Referee #2 (Remarks to the Author):

Building on their previous work, the authors provide structural and biochemical evidence that Omicron evades antibodies targeting the S2'-helix through two primary mechanisms: reducing S2'-helix exposure while accelerating spike refolding and shifting viral entry toward a CTSL-dependent pathway. The study presents high-resolution cryo-EM structures of spike refolding intermediates transitioning from the prefusion to postfusion state, which is valuable for understanding spike dynamics and informing future antibody design.

### Response:

Thanks for the positive comments on our manuscript.

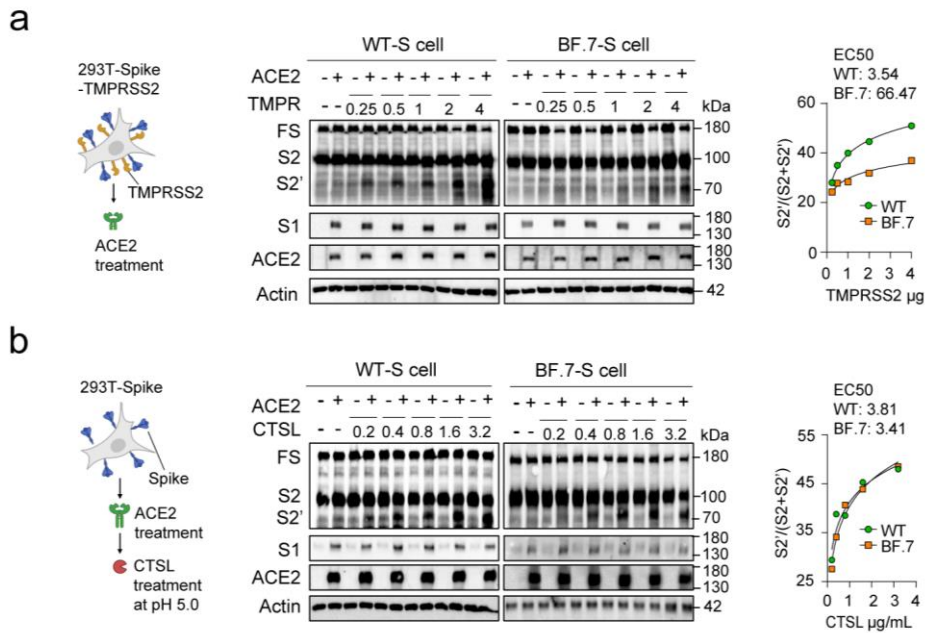
1, However, some conclusions remain speculative without direct experimental support. For instance, the impact of TMPRSS2 and CTSL size, shape, and binding orientation on S2-helix accessibility and subsequent cleavage by neutralizing antibodies is not addressed.

### Response:

We sincerely appreciate your valuable comments, which are crucial for improving the accuracy and significance of our manuscript. We apologize for our insufficiently precise description regarding this issue.

For the concern “the impact of TMPRSS2 and CTSL size, shape, and binding orientation on S2-helix accessibility”, we have ever attempted to determine the cryo-EM structures of the inactivated TMPRSS2 or CTSL proteins complexed with ACE2 and S proteins, respectively. Unfortunately, these attempts were unsuccessful, which may be due to weak binding affinity between the proteases and the spike protein, as the demanding sample preparation conditions for cryo-EM could readily disrupt weakly associated complexes. Notably, our search of the entire PDB database revealed no existing structures of any coronavirus spike protein in complex with TMPRSS2 or CTSL. However, for the “subsequent cleavage of TMPRSS2 and CTSL”, we have clearly observed distinct cleavage capabilities of TMPRSS2 and CTSL towards the S2’ site in both WT and Omicron S proteins. Omicron BF.7 S protein showed reduced reliance on TMPRSS2 for cleaving the S2’ site. In contrast, CTSL cleaves both WT and Omicron S proteins with comparable efficiency following ACE2 priming (**Fig. 3a-b**), which may contribute to the altered cellular entry pathway preference observed in Omicron variants.

To reflect this understanding and ensure clarity, we have deleted the original descriptions and revised the relevant descriptions shown as above in the *Main text* (**Lines 250-257**), and *Discussion* (**Lines 399-404**).



**Fig. 3 Increased utilization of CTSL-dependent entry pathway of Omicron enables evasion from 76E1. a-b,** Dose-dependent cleavage of cell-expressed WT or Omicron BF.7 S proteins at the S2' site by TMPRSS2 (a) or CTSL (b), with or without ACE2-Fc, assessed by Western blot. **a,** 293T-S cells were transiently transfected with TMPRSS2 for 24 h, followed by 2 h ACE2 treatment to initiate cleavage. **b,** CTSL cleavage was performed at pH 5.0 for 20 min after ACE2 pre-incubation. Cleavage EC<sub>50</sub> values were calculated from band intensity ratios (S2'/(S2'+S2)). representative of  $\geq 2$  experiments.

2, Additionally, S1 shedding in response to ACE2 treatment and the effect of antibodies on membrane fusion should be compared, given the role of ACE2 priming in S2 cleavage and fusion activation. Quantification of protein band intensities would further strengthen the conclusions.

3, Fig. 1E and other Western blots: Actin and spike signal variations across samples necessitate the tested antibody input control. S1 shedding should be shown, and band intensities quantified. Clarification on whether WT refers to the ancestral Wuhan strain or D614G is needed.

4, Fig. 3: It's important to quantify S and S2' band intensities and demonstrate dose-dependent effects of TMPRSS2, CTSA, and antibody treatments, with IC<sub>50</sub>

calculations for WT and BF.7. Include shed S1 as a control to compare ACE2 priming effects.

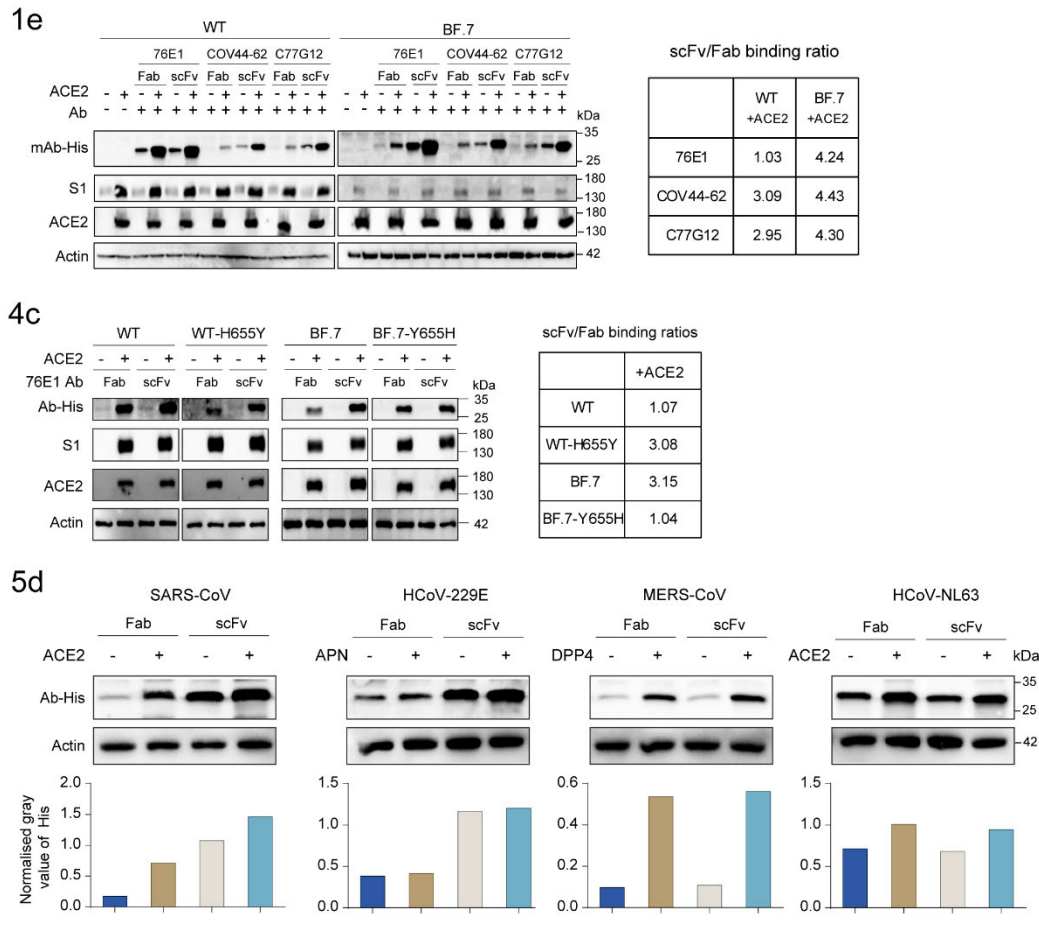
**Response to 2-4:**

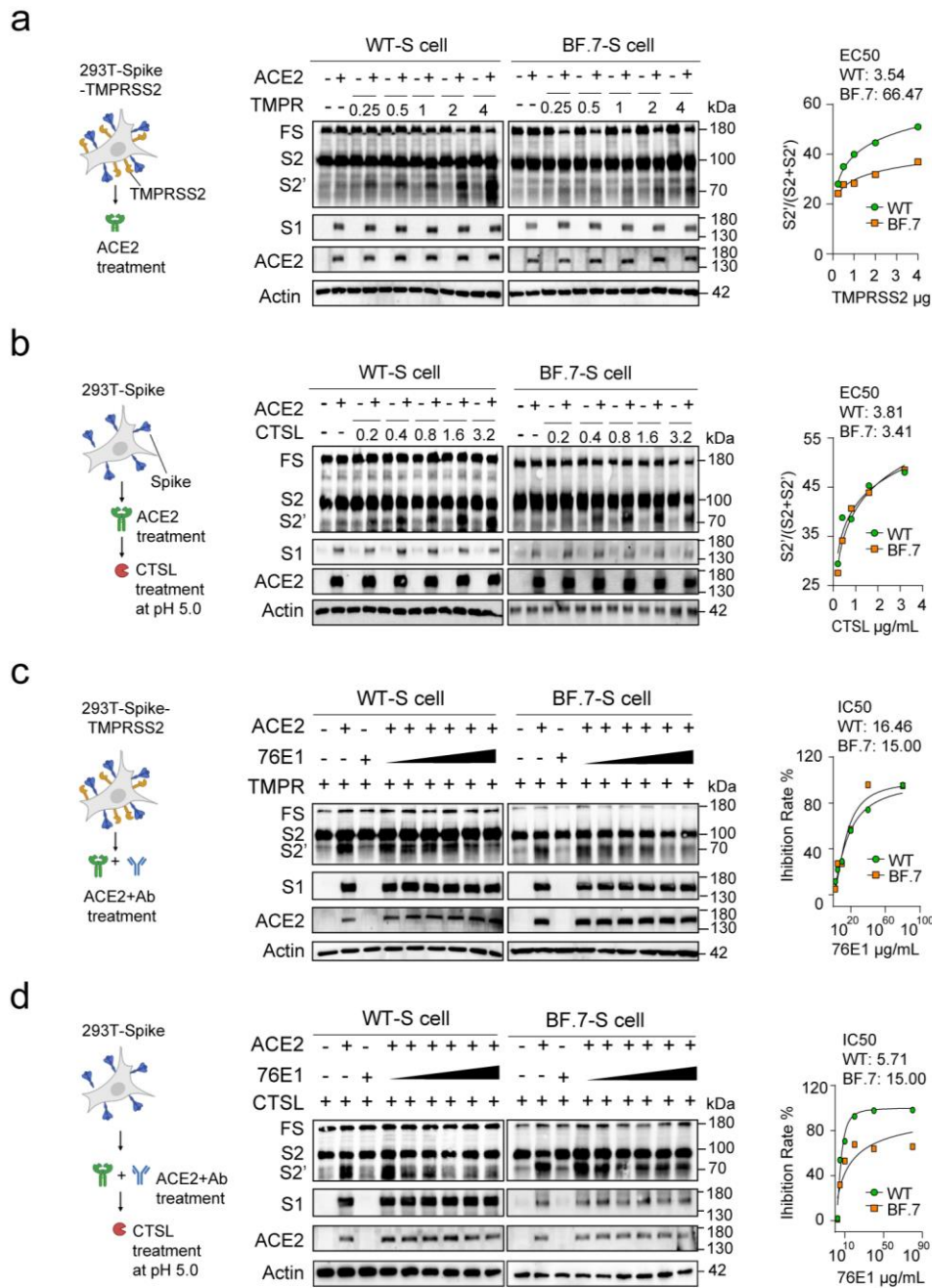
We thank the reviewer for the valuable suggestion. For the reviewer's comparable comments above, we put together a response. Please find our point-by-point reply below:

(1) To quantitatively analyze all Western blot (WB) data, dose-dependent WB experiments were performed. As it is hard to achieve identical protein expression levels between the two S-expressing cells, we performed independent analyses for each spike variant. All comparisons were normalized through quantitative assessment and calculation of band intensities, including: ① Antibody binding ratios (scFv/Fab): calculated from scFv-his and Fab-his band intensities (**Fig. 1e, 4d, 5d**) to assess enhanced spike-binding activity after antibody size reduction, indicating potential steric hindrance; ② S2' cleavage proportion ( $S2'/(S2'+S2)$ ): calculated from S2' band intensities (**Fig. 3a-b**) to measure TMPRSS2- or CTSL-mediated cleavage at the S2' site; ③ S2' cleavage reduction rates: quantified in 76E1-treated groups (**Fig. 3c-d**) to determine 76E1's inhibitory efficiency on TMPRSS2- or CTSL-mediated S2' cleavage.

(2) S1 shedding served as the sample control in all WB experiments.

(3) In addition, we have clarified that the WT refers to the ancestral Wuhan strain in the *Main text* as the reviewer suggested. (**Line 101**)





**Fig. 3 Increased utilization of CTSL-dependent entry pathway of Omicron enables evasion from 76E1. a-b,** Dose-dependent cleavage of cell-expressed WT or Omicron BF.7 S proteins at the S2' site by TMPRSS2 (a) or CTSL (b), with or without ACE2-Fc, assessed by Western blot. **a,** 293T-S cells were transiently transfected with TMPRSS2 for 24 h, followed by 2 h ACE2 treatment to initiate cleavage. **b,** CTSL cleavage was performed at pH 5.0 for 20 min after ACE2 pre-incubation. Cleavage EC<sub>50</sub> values were calculated from band intensity ratios ( $S2'/(S2'+S2)$ ). **c-d,** Dose-dependent inhibition of TMPRSS2 (c) or CTSL (d)-mediated S2' cleavage by 76E1-FL. Cleavage inhibition IC<sub>50</sub> values were derived from S2' band intensities normalized to no-antibody controls.

representative of  $\geq 2$  experiments.

**5, A high-resolution structure of XBB.1.5 or JN.1 spike in complex with E-FIC-76E1 would also be informative, considering similarities in 2D classification and 3D reconstruction between WT and XBB.1.5 L-FIC-76E1.**

**Response:**

Thanks for the insightful suggestion. We optimized the sample preparation of XBB.1.5 S<sup>E-FIC</sup>-76E1 complex and determined the structure to an overall resolution of 5.65 Å. The S2-76E1 and S2-76E1 top regions were locally refined to 4.14 Å, allowing a structural analysis. (**Extended Data Fig. 5**). Additional details on sample preparations, cryo-EM data collections and processing, and modeling are included in the *Methods* section in the revised manuscript.

Based on the newly resolved structure of XBB.1.5 S<sup>E-FIC</sup>-76E1 and our previously reported structure of apo D614G S<sup>E-FIC</sup>, we reorganized the figures (**Fig. 2 and Extended Data Fig. 3-7**) and rewrote the manuscript. The related descriptions are included into the *Cryo-EM structures of WT intermediate S recognized by 76E1-Fab* and *Cryo-EM structures of XBB.1.5 intermediate S recognized by 76E1-Fab* sections of the revised manuscript. (**Lines 195-246**)

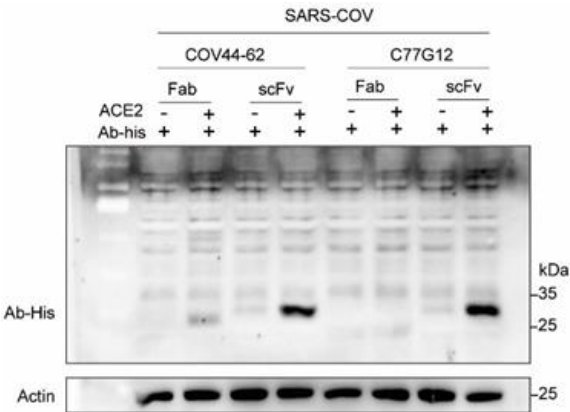
**Specific comments:**

**6, Neutralization and structure: Assessing COV44-62 and C77G12 binding to the SARS-CoV S2'-helix will be helpful.**

**Response:**

We sincerely appreciate your valuable comments, and your concern about the binding accessibility of COV44-62 and C77G12 to SARS-CoV S2'-helix is also what we care about. To address this issue, we constructed two his-tagged antibody formats varying in molecular size: the Fab fragment (~50 kDa) and scFv fragment (~27 kDa). Cell surface-expressed SARS-CoV S proteins were treated with these antibody formats at equivalent molar concentrations after ACE2 priming. Both COV44-62 and C77G12

scFv exhibited enhanced binding activity to SARS-CoV S than Fab (Fig. R1). These findings indicate that the SARS-CoV S is likely to have more steric hindrance to the large-sized COV44-62 and C77G12 Fab, which is similar to the Omicron S. Therefore, reducing antibody molecular size remains an effective strategy to enhance neutralizing potency against SARS-CoV, attributable to analogous steric constraints.



**Fig. R1** Western blot analysis of COV44-62 and C77G12 scFv and Fab binding activity at equal molar concentrations to cell-surface-expressed SARS-CoV spike proteins, with or without ACE2-Fc.

7, The effects of E64d and Camostat appear modest, and the Delta strain, which primarily utilizes plasma membrane entry, could serve as a control.

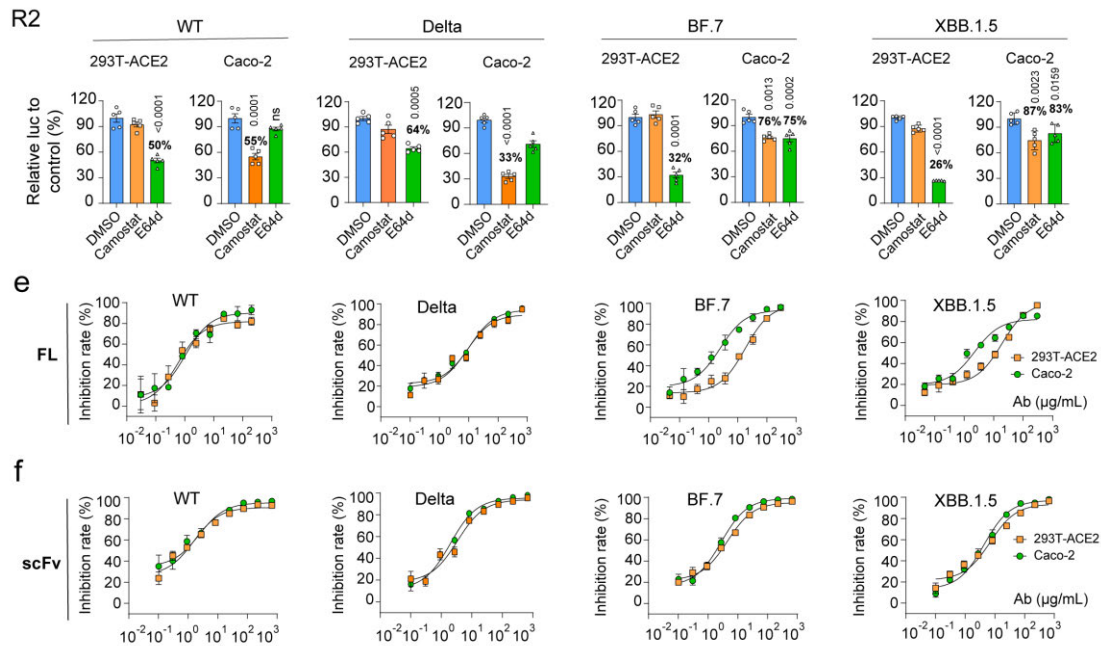
**Response:**

We thank the reviewer for the valuable suggestion. Please find our point-by-point reply below:

(1) Regarding “the modest effects of E64d and Camostat”, we note Bingjie Hu et al. reported  $\leq 50\%$  viral entry inhibition using 25-50  $\mu\text{M}$  E64d in Vero-E6 cells (endosomal pathway-dominant) and  $\sim 50\%$  inhibition with equivalent Camostat doses in Calu-3 cells (plasma membrane pathway-dominant). The inhibitors in this paper were purchased from MedChemExpress<sup>2</sup>. In contrast, Hoffmann et al. demonstrated near-complete ( $\approx 100\%$ ) inhibition at 25  $\mu\text{M}$  E64d in 293T-ACE2 cells (endosomal entry-dominant) and Camostat in Caco-2 cells (surface fusion-dominated). The

inhibitors in this paper were purchased from Sigma-Aldrich<sup>3</sup>. Separately, ~70% inhibition in Caco-2 cells was observed with 1-25  $\mu$ M Camostat (Sigma-Aldrich)<sup>4</sup>. In our study, 10  $\mu$ M E64d or Camostat (Selleck) achieved approximately 50% viral entry inhibition in both 293T-ACE2 and Caco-2 models. These divergent efficacies across studies likely stem from differential inhibitor concentrations, potential batch variations between suppliers, and cell line-specific responses. Generally, the drug dosages administered in our study reside within the reasonable range, and the observed inhibitory trends closely paralleled those documented in prior reports.

(2) Additionally, in accordance with the reviewer's suggestion, we repeated the experiments and “included the Delta variant as an additional control”. Consistent with the WT strain, Delta exhibited greater sensitivity to Camostat in Caco-2 cells compared to Omicron subvariants BF.7 and XBB.1.5. Conversely, both Omicron strains demonstrated enhanced sensitivity to E64d in 293T-ACE2 cells relative to Delta (**Fig. R2**). These findings align with established reports that WT and Delta primarily utilize plasma membrane fusion for cellular entry, whereas Omicron variants preferentially employ the endosomal pathway, demonstrating the technical reliability of our experimental system. Furthermore, we incorporated Delta as a control in **Fig. 3f-g** to evaluate neutralizing activities of 76E1-scFv and full-length antibodies across 293T-ACE2 and Caco-2 cell lines. The data recapitulated the pattern observed with WT: both FL and scFv antibody formats neutralized Delta with comparable efficacy in both cellular models (**Fig. 3e-f**). The related descriptions are included in the *Increased utilization of CTSL-dependent entry pathways underlies Omicron immune evasion from 76E1* section. (Lines 258-275)



**Fig. R2.** Relative pseudovirus infection of SARS-CoV-2 WT, Delta and Omicron variants BF.7 and XBB.1.5 (compared to DMSO-treated) after 10  $\mu$ M protease inhibitor- Camostat or E64d treatment in 293T-ACE2 or Caco-2 cell lines, respectively.  $n=3$ . Data are shown as mean  $\pm$  SEM. Two-sided unpaired t-tests,  $P > 0.05$  (non-significant, *n.s.*),  $*P \leq 0.05$ ,  $**P \leq 0.01$ ,  $***P \leq 0.001$ ,  $****P \leq 0.001$ .

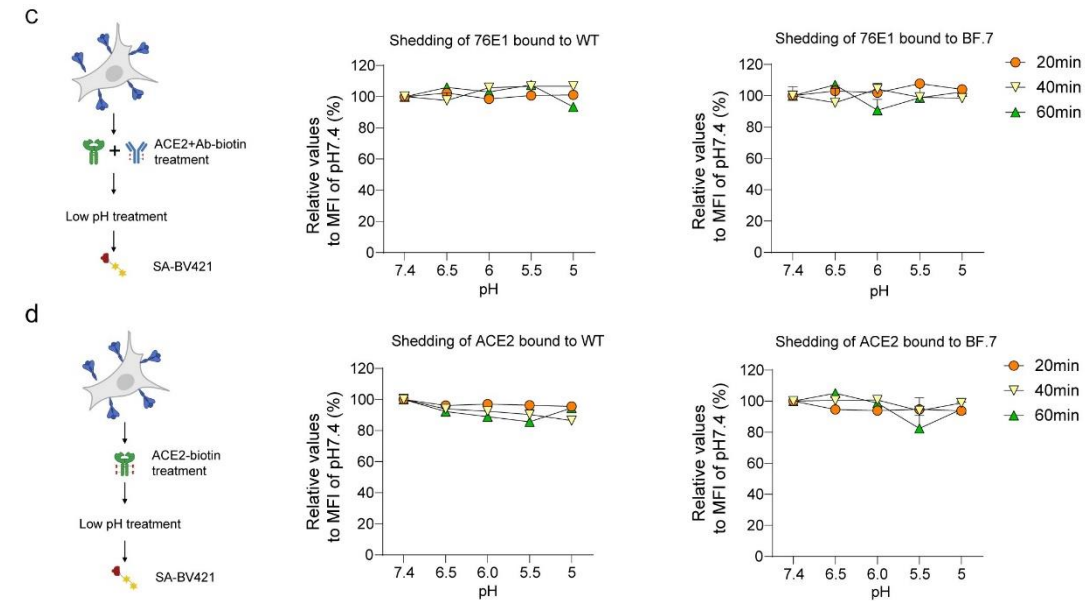
**Fig. 3e-f,** Neutralization curves of 76E1-FL (**e**) and scFv (**f**) against SARS-CoV-2 pseudoviruses (WT, Delta, BF.7, XBB.1.5) in 293T-ACE2 and Caco-2 cells, respectively.  $n=3$ . Data: mean  $\pm$  SEM; representative of  $\geq 2$  experiments.

**8, Acidic environment and antibody escape:** It will be important to test Ab binding and shedding over time under different pH conditions at 37°C, rather than 20 min, to substantiate the claim that acidic pH does not affect on Ab shedding.

## Response:

We thank the reviewer for this valuable suggestion. In response, we repeated the FCM assays with extended incubation periods (20, 40, and 60 min) under varying pH conditions at 37 °C. Our data demonstrate that exposure of 76E1-bound 293T-BF.7 S cells to low-pH environments, even for 60 min, fails to significantly induce shedding of either 76E1 antibody or ACE2 (**Extended Data Fig. 8c-d, Supplementary Fig. 5**). Consequently, these findings indicate that endosomal acidity does not represent a major

mechanism underlying Omicron’s evasion of S2’-helix-targeting antibodies. The related descriptions are revised in the *Main text* and *Methods* sections. (Lines 276-289)



**Extended Data Fig. 8c-d c**, Exposure of 293T-S cells that bound with the 76E1-ACE2 complex to low pH conditions did not significantly elicit shedding of the 76E1 antibody. 76E1 and ACE2 pre-treated 293T-S cells were subjected to various low pH buffers. Following exposure, cells retaining 76E1 antibody binding after washing were stained with streptavidin-conjugated BV421 and subsequently analyzed via FACS. **d**, Exposure of ACE2-bound 293T-S cells to low pH conditions does not significantly induce ACE2 shedding. The 293T-S cells were treated with ACE2-biotin protein and subsequently detected using streptavidin-conjugated BV421, followed by analysis via FACS. Relative MFI values of each low pH compared to pH 7.4 were shown.  $n=3$ . Data are shown as mean  $\pm$  SEM. For all data, one representative experiment of at least two is shown.

**9, H655Y mutation and entry pathway shift:** The assertion that H655Y shifts viral entry preference to the endosomal route by decreasing S2’-helix exposure requires additional experimental validation. Reversion key mutants in Omicron should be tested for entry pathway and S2’-helix exposure.

**Response:**

Thanks for the reviewer’s valuable suggestion, which are important to improve the significance of our data. We reverted H655 to Y655 mutation in Omicron BF.7 S and

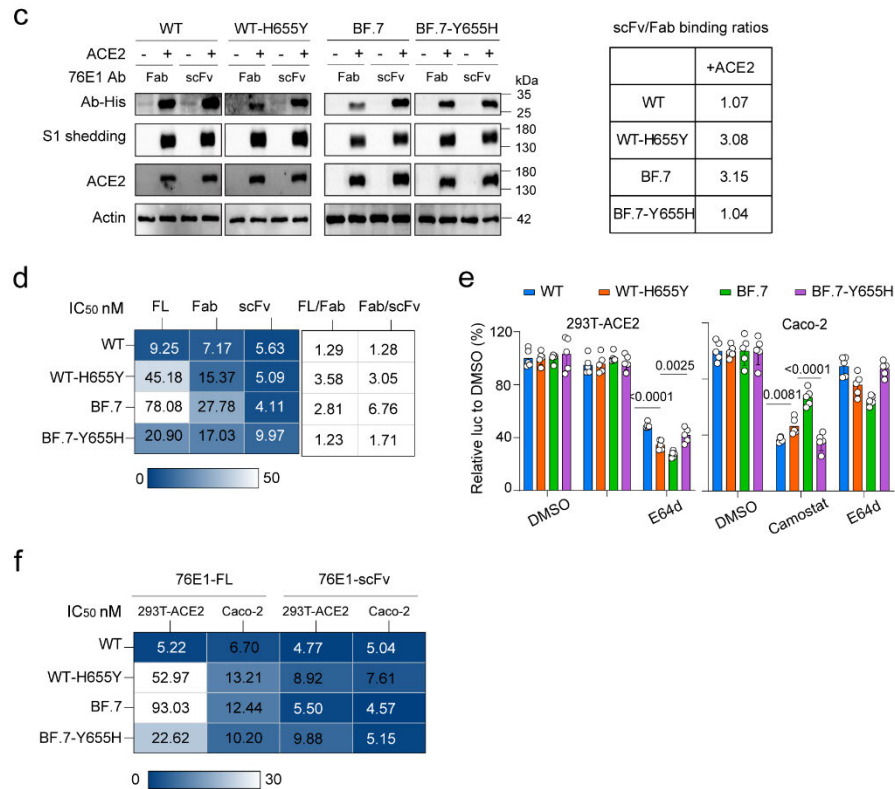
constructed BF.7-Y655H S expressing 293T cells and pseudovirus for further study.  
BF.7 and WT groups were included as the controls.

The Y655H reversion mutation in BF.7 S diminished steric hindrance at the S2'-helix epitope, as evidenced by reduced disparity in binding between 76E1-scFv and Fab relative to parental BF.7 (**Fig. 4c**). Consistent with the binding data, neutralization assays in 293T-ACE2 cells showed that the reversion mutation of Y655H in the S protein of Omicron BF.7 narrowed the neutralizing gap between 76E1-FL and Fab (**Fig. 4d**). These findings further indicate that H655Y mutation hinder direct antibody access to the S2'-helix.

For the entry pathway related assays, BF.7- Y655H pseudovirus showed decreased dependency on endosomal entry pathways compared with parental BF.7 pseudovirus (**Fig. 4e**), which is correlated to the data on WT-H655Y.

Subsequent evaluation of the neutralizing activity of 76E1-FL in targeted cells with distinct viral entry pathways indicated that Y655H reversion mutation in BF.7 spike reduced the neutralization differences across cell lines for 76E1-FL (**Fig. 4f**), which is also correlated to the data on WT-H655Y.

The related descriptions are included in the *The H655Y mutation primarily enables Omicron variants to escape 76E1* section. (Lines 300-332)



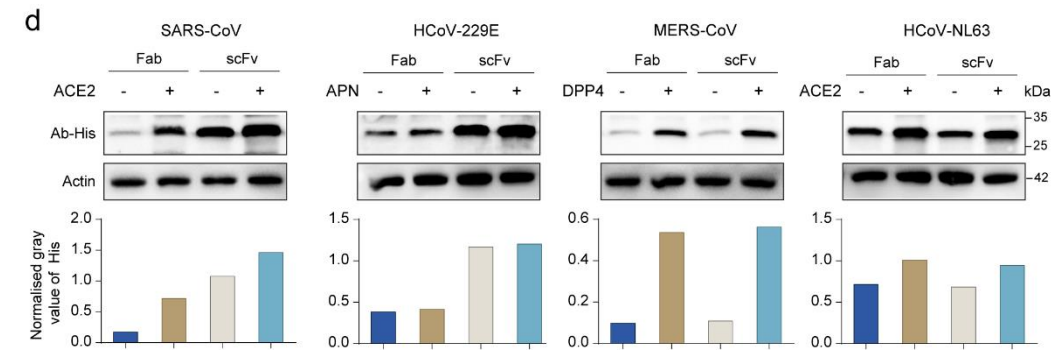
**Fig. 4 The H655Y mutation primarily enables Omicron variants to escape 76E1.** **c**, Western blot analysis of 76E1-scFv and Fab binding at equal molar concentrations to cell-surface-expressed WT-H655Y S proteins, with or without ACE2-Fc. WT, BF.7 and reverted mutation of Y655H in BF.7 as controls. His-tagged antibody formats were detected using anti-His-HRP antibody. S1 subunit shedding into the cell culture supernatant was quantified. ScFv/Fab ratios post-ACE2 pre-incubation calculated from band intensities. **d**, Neutralizing IC<sub>50</sub> of 76E1-FL, Fab and scFv at equal molar against indicated pseudoviruses in 293T-ACE2 cells. Data are shown as mean  $\pm$  SEM. **e**, Relative infection of H655Y pseudovirus (compared to DMSO-treated group) post 10 $\mu$ M Camostat/E64d treatment in 293T-ACE2 or Caco-2 cells. WT, BF.7 and BF.7-Y655H were included as controls.  $n=5$ . Data are shown as mean  $\pm$  SEM. Two-sided unpaired t-tests,  $P > 0.05$  (non-significant, n.s.),  $*P \leq 0.05$ ,  $**P \leq 0.01$ ,  $***P \leq 0.001$ ,  $****P \leq 0.0001$ . **f**, Neutralizing IC<sub>50</sub> values of 76E1-FL and scFv in 293T-ACE2, as compared with Caco-2 cells.  $n=3$ . Data are shown as mean  $\pm$  SEM.

**10, Comparative spike refolding exposure:** Experimental evidence should support claims that SARS-CoV-1 and HCoV-229E spikes exhibit less exposed S2-helix during refolding compared to MERS-CoV and HCoV-NL63, particularly in response to full-

length and scFv antibodies.

**Response:**

We appreciate the reviewer's insightful inquiry, which aligns directly with our research focus. To investigate the accessibility of the S2'-helix epitope to 76E1 antibody across diverse human coronaviruses and evaluate potential steric hindrance effects, we generated spike protein stably expressed 293T cell lines of *alphacoronaviruses* HCoV-NL63, HCoV-229E and *betacoronaviruses* MERS-CoV, SARS-CoV-1. Following priming with cognate receptors, ACE2 for SARS-CoV-1 and HCoV-NL63, APN for HCoV-229E, and DPP4 for MERS-CoV, the cells were treated with 76E1-scFv or Fab fragments. Antibody binding was assessed via immunoblotting of the His-tag. Notably, 76E1-scFv exhibited enhanced binding to SARS-CoV-1 and HCoV-NL63 S compared to 76E1-Fab, indicating significant steric hindrance at the S2'-helix epitope of 76E1-Fab than scFv (**Fig. 5d**). Conversely, for HCoV-229E and MERS-CoV S, comparable binding between scFv and Fab formats suggests better epitope exposure and equivalent accessibility. The related descriptions are included in the *Minimization of 76E1 antibody enhanced the neutralizing activity against authentic Omicron subvariants and some other human coronaviruses* section. (Lines 356-358)



**Fig. 5d.** Western blot analysis of 76E1 scFv and Fab binding at equal molar concentrations to cell-surface-expressed spike proteins of indicated coronaviruses, with or without receptor pre-incubation. Top: Anti-His-HRP detection; Bottom: Band intensity quantification. One representative experiment of at least two is shown.

### Referee #3 (Remarks to the Author):

In this manuscript, the authors describe how the format/size of S2' helix antibodies modulate their neutralization potency in various context and they obtained a cryogenic-electron microscopy structure of a SARS-CoV-2 spike fusion intermediate with the 76E1 Fab. The results are interesting although not all conclusions are supported by the data and some of the results appear to overlap with an article published in Cell by some of the same authors (PMID: 39889696) .

### Response:

We sincerely appreciate the reviewer's insightful comments, which have significantly contributed to improve the quality of our work. In the revised manuscript, we have incorporated additional experimental data and refined textual descriptions to better support our conclusions.

In response to the concern regarding potential overlap with our previously published paper in Cell<sup>1</sup> (Cell, 2025, PMID: 39889696), we conducted a detailed comparative analysis and highlighted the distinct conceptual advances of the present study and emphasized the novel insights it provides. Specifically, the main contributions of this work are summarized as follows:

**(1) We provide mechanistic insights into the neutralization activity of S2'-helix-targeting antibody by resolving the structure of 76E1 in complex with SARS-CoV-2 spike fusion intermediate.** Specifically, 76E1 recognizes an early fusion intermediate conformation (E-FIC) of the S protein that is more open than the previously reported apo D614G S<sup>E-FIC</sup> structure (PDB 8Z7P). In the WT S<sup>E-FIC</sup>-76E1 complex, three 76E1-Fabs are embedded into the E-FIC in a form parallel to the ACE2s' inner surface, binding to a S2'-helix (residues 815-825) in IL770 of each S protomer (**Fig. 2a-c**).

The epitope is conformationally flexible in apo D614G S<sup>E-FIC</sup> and is poorly characterized. However, upon 76E1 binding, this region becomes rigid and well-

resolved (**Extended Data Fig. 6g-k**). By engaging this opened E-FIC, 76E1 sterically blocks access to the S2' cleavage site and thereby inhibits the subsequent refolding of S2 into the postfusion state. These findings are presented in the revised manuscript under the section titled the *Cryo-EM structures of WT intermediate S recognized by 76E1-Fab*.

**(2) We elucidate the mechanism by which 76E1 is evaded by Omicron variants through functional and structural analysis.** Using optimized sample preparation procedures (as described in the *Methods* section), we determined the cryo-EM structure of XBB.1.5 S<sup>E-FIC</sup>-76E1 (**Fig. 2a and Extended Data Fig. 5**).

Structural comparison with WT S<sup>E-FIC</sup>-76E1 revealed Omicron-specific conformational rearrangements in the S2-top region containing the S2'-helix epitope. In XBB.1.5 S<sup>E-FIC</sup>-76E1, the S2-top rotates counterclockwise by ~6.4° around the threefold axis, and the S2'-helix itself is displaced downward by ~1.2 Å. In contrast, the S1-ACE2s units are shifted upward by ~2.6 Å. These combined shifts reduce the available space for 76E1 binding and introduce steric hindrance against the antibody (**Fig. 2d-i and Extended Data Fig. 7**). To accommodate 76E1, the S1-ACE2s undergo a counterclockwise rotation of ~3.8°, with ACE2s further rotating outward ~2.2°. These findings highlight the conformational plasticity of the Omicron spike, where subtle yet coordinated structural rearrangement enables immune evasion from S2'-targeting antibodies. The related descriptions have been included in the *Cryo-EM structures of XBB.1.5 intermediate S recognized by 76E1-Fab* section of the revised manuscript.

**(3)** Additionally, reduced TMPRSS2-mediated S2' cleavage efficiency in Omicron than WT, contrasted with potent CTSL cleavage of both, possibly explains Omicron's altered preference to CTSL-dependent entry pathway, thereby contributing to 76E1's diminished efficacy through epitope inaccessibility. The related descriptions are included into the *Increased utilization of CTSL-dependent entry pathways underlies Omicron immune evasion from 76E1* section of the revised manuscript.

**(4)** We identify a key mutation, H655Y, that mediates immune evasion of

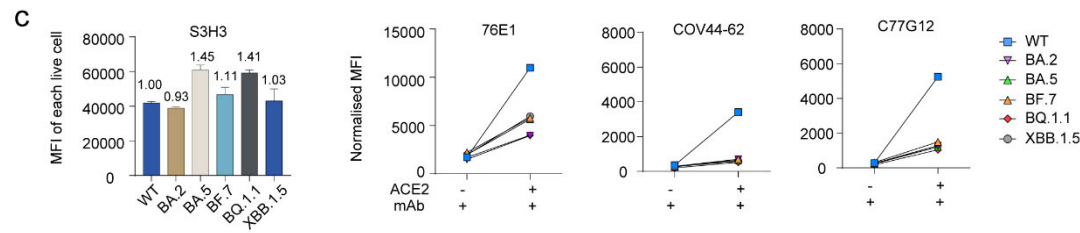
Omicron through dual mechanisms: enhanced steric hindrance at the S2'-helix and switch preference to CTSL-dependent entry pathways during infection.

(5) Minimized antibody formats targeting the S2'-helix significantly enhance neutralization of Omicron and some other human coronaviruses including SARS-CoV-1 and HCoV-229E. We therefore propose small-molecule inhibitors targeting this epitope as a viable broad-spectrum strategy.

1, Lines 113-117: Have the authors controlled for differences of WT vs XBB spike expression?

**Response:**

We thank the reviewer for this valuable suggestion. Given the challenge in achieving absolute equivalence of S protein expression in 293T cells between WT and Omicron (**Fig. 1c**), we implemented a quantitative normalization strategy. Specifically, we assessed S3H3 antibody binding via FCM across all S-expressing cell lines. S3H3, an SD1-targeting antibody with documented comparable neutralization against WT and Omicron subvariants including BA.2, BA.4, BA.5, BQ.1.1, and XBB.1.5<sup>5</sup>, served as an internal reference for spike expression levels. We calculated the relative S3H3 binding activity of BA.2, BA.4, BA.5, BQ.1.1 and XBB.1.5 to WT, with WT as 1.00, and BA.2, BA.4, BA.5, BQ.1.1 and XBB.1.5 were then calculated as 0.93, 1.45, 1.11, 1.41 and 1.03, respectively. Following normalization to these S3H3 values, 76E1, COV44-62, and C77G12 binding profiles in **Fig. 1c** demonstrated significantly reduced binding to ACE2-primed Omicron spikes than WT. We have updated **Fig. 1c** and refined corresponding descriptions in the *Main text* and **Figure 1 legend** to reflect this normalization approach. (Lines 115-121)



**Fig. 1c**, Binding of full-length 76E1 (76E1-FL) to cell-surface-expressed spike proteins of SARS-

CoV-2 variants, with or without ACE2-Fc, measured by flow cytometry. S3H3, an SD1-targeting antibody used to quantify and normalize for spike expression levels across cell lines due to its comparable neutralization against variants. Relative S3H3 MFI binding ratios to WT spike are shown in the left panel. The right three panels show normalized MFIs of 76E1, COV44-62, and C77G12 relative to S3H3.  $n=3$ . One representative experiment of at least two is shown.

**2, Lines 124-130: How do the authors explain the strong scFv binding of COV44-62 and C77G12 in the absence of ACE2 (Figure 1e)?**

**Response:**

Regarding the scFv binding of COV44-62 and C77G12 in the absence of ACE2, we conducted extensive literature analysis. Hangping Yao et.al reported needle-like densities were occasionally observed on the inactivated SARS-CoV-2 viral envelope by cryo-electron tomography (cryo-ET) and subtomogram averaging (STA), suggesting a postfusion conformation of the S protein<sup>6</sup>. Moreover, Yongfei Cai et.al reported cryo-EM structures derived from a preparation of the full-length S protein, representing its prefusion and postfusion conformations, respectively. The spontaneous transition to the postfusion state is independent of target cells<sup>7</sup>. Additionally, spontaneous S1 shedding from SARS-CoV-2 pseudoviruses in the absence of ACE2 has been reported<sup>8</sup>. These findings collectively indicate the existence of ACE2-independent conformational dynamics in the spike protein. While such intrinsic conformational plasticity likely operates at low basal levels, ACE2 binding substantially amplifies the structural transition to postfusion state. Therefore, we hypothesized that even in the absence of ACE2, conformational rearrangement of the S protein that exposes the S2'-helix epitope might enable direct 76E1-scFv binding.

**3, Lines 136-139: How do the authors reconcile the conclusion that Omicron variants reduce S2' helix epitope exposure given that they were able to image a stalled E-FIC for both WT and XBB.1.5 spikes? Are the cryoEM structures suggesting an explanation for this mechanism?**

**Response:**

We sincerely appreciate the reviewer's insightful comments, which have been pivotal in refining our manuscript. To more precisely characterize the reduced antibody accessibility to the Omicron S2'-helix epitope relative to wild-type (WT), we resolved the XBB.1.5 S<sup>E-FIC</sup>-76E1 structure using the optimized sample preparation methods described earlier. Furthermore, we have revised the corresponding text in the manuscript, replacing the phrase "decreased epitope exposure" with "enhanced steric hindrance" to more precisely reflect the structural basis of antibody evasion in Omicron.

This conclusion is supported by two key observations:

**(1) Antibody size-dependent neutralizing activity:** Reduction of antibody size (e.g., conversion of 76E1 to Fab and scFv format) significantly enhanced binding and neutralizing activities against Omicron variants compared to WT and Delta. These data conclusively demonstrate that antibody size directly modulates their accessibility to this epitope. Omicron displays greater steric occlusion at its S2' site than WT, which directly affect antibody binding to this epitope.

**(2) Comparative structural analysis:** To clarify the reduced antibody accessibility in Omicron, we performed a comparative structural analysis between WT S<sup>E-FIC</sup>-76E1 and XBB.1.5 S<sup>E-FIC</sup>-76E1 complexes. In XBB.1.5 S<sup>E-FIC</sup>-76E1, the S2-top region containing the S2'-helix epitope is rotated counterclockwise by approximately 6.4° around the 3-fold axis and the S2'-helix moves downward by ~1.2 Å. In contrast, the S1-ACE2s move upward by ~2.6 Å, resulting in a narrower interfacial space and introduces steric hindrance that impedes 76E1 binding (**Fig. 2d-i and Extended Data Fig. 7**). To accommodate 76E1 under these conformational constraints, the S1-ACE2s undergo an additional counterclockwise rotation (~3.8°), with ACE2 itself rotating outward by ~2.2°, thereby requiring increased energetic input for binding.

These observations indicate that the physical gap between ACE2s and the S2'-helix epitope in E-EIC conformation is a critical determinant of antibody accessibility. Therefore, both the antibody size and conformational changes have the potential to influence the accessibility of the epitope.

4, Lines 149-151: Please rephrase as it is simply that 76E1 does not bind prefusion spike.

**Response:**

Thanks for the suggestion. We have rephrased these statements as “*76E1 cannot bind to prefusion S due to the partial burial of its epitope*” in our revised manuscript.

5, Lines 155-157: How do the authors know that the reconstruction obtained correspond to the L-FIC considering the severe preferred orientation and associated low resolution?

[REDACTED]

6, Lines 159-163: CH does not change conformation relative to prefusion spike.

**Response:**

Thanks for pointing this out. We have corrected it in the revised manuscript.

7, Lines 186-188: Has a control experiment been carried out without 76E1 and was post fusion observed? The authors claim that Omicron spike refold more rapidly than WT spike. This statement is not supported by any data on refolding kinetics.

8, Lines 198-200: It suggests reduced stability (delta G) not faster refolding kinetics of individual spikes. Also, how do the authors reconcile this with the presence of D614G in XBB.1.5 but not WT which is supposed to enhance prefusion stability?

9, Have the authors tried adding 76E1-scFv and did this prevent transition to the postfusion spike?

**Response to 7-9:**

Thanks for pointing out these issues and providing instructive suggestions. In response to the reviewer's comments (7-9), we have made the following revisions:

(1) As suggested, we have removed this inappropriate statement about "accelerated spike refolding" in the revised manuscript.

(2) As mentioned above, we evaluated the structural transitions of WT, D614G and XBB.1.5 S in the presence of 76E1-Fab and scFv after ACE2 priming using negative staining electron microscopy. Overall, XBB.1.5 S demonstrated reduced stability relative to the WT and D614G S while ACE2-activated D614G and XBB.1.5 S rapidly adopted a postfusion conformation compared to WT S. Meanwhile, few discernible postfusion particles were observed in the mixtures containing 76E1-Fab and scFv, suggesting both 76E1-Fab and scFv could inhibit the transition to the postfusion state. These findings are now incorporated into the revised manuscript (**Supplementary Fig. 1-3**):

*"The 76E1-Fabs could effectively block the transition of ACE2-primed WT S to the postfusion conformation, as no postfusion particles were observed in the WT S-ACE2-76E1 mixture, even after 120 min incubation, in contrast to the WT S-ACE2 mixture (Supplementary Fig. 1)." (Lines 185-189)*

832 “XBB.1.5 S demonstrated reduced stability relative to the WT and D614G S, while  
833 ACE2-activated D614G and XBB.1.5 S rapidly formed a postfusion conformation  
834 (**Supplementary Fig. 1-3**). Few discernible postfusion particles were observed in the  
835 mixture containing 76E1-Fab and scFv.” (Lines 197-201)

836 **10**, Lines 201-210: What is the molecular rational for this? Were the native spike ECDs  
837 co-expressed with furin (as recently done by the same authors, PMID: 39889696)? If  
838 not, what was the rational to employ a different approach?

839 **Response:**

840 We thank the reviewer for pointing out these important issues about the molecular  
841 rationality. Please find our point-by-point reply below:

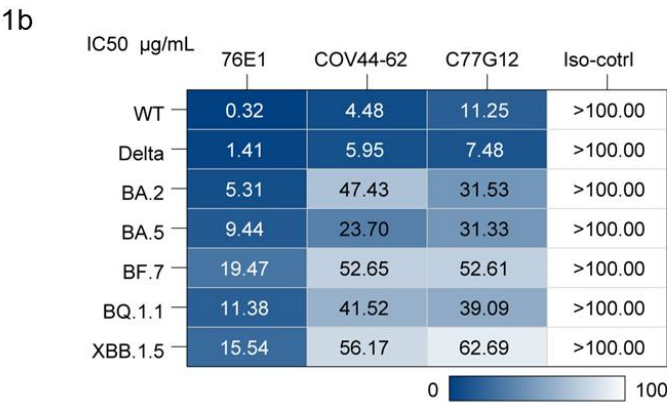
842 (1) Response to the issue “**Were the native spike ECDs co-expressed with furin...**”:  
843 In this study, we expressed native spike ectodomains (ECDs) without furin co-  
844 expression. We observed that ACE2 binding was able to trigger the prefusion-to-  
845 postfusion conformational transition of the purified S trimer, containing mainly the  
846 cleaved S. While furin co-expression may accelerate this structural transition, it could  
847 also exaggerate the instability of certain spike variants, such as Omicron. As furin-free  
848 preparations adequately supported our structural and functional analyses, we utilized  
849 this simplified system throughout our experiments.

850 (2) Response to “**What is the molecular rational for (Lines 201-204)?**”  
851 (**Conversely, the increase in postfusion particles suggests that 76E1 cannot fully inhibit**  
852 **the conformational transition of the spike protein from prefusion to postfusion states,**  
853 **which explains its partial evasion observed in neutralization assays (Fig. 2a and Fig.**  
854 **1b) (Lines 201-204))**

855 We appreciate the reviewer’s comment and acknowledge that the original  
856 statement was inappropriately phrased. In the revised manuscript, we have reworded  
857 the sentence as follows:

858 “The increase in postfusion particles suggests that 76E1 cannot fully inhibit the

conformational transition of the XBB.1.5 S from prefusion to postfusion states, which is consistent with its partial evasion observed in neutralization assays (Fig. 1b).”



**Fig.1b** The neutralizing IC<sub>50</sub> values of full-length 76E1, COV44-62 and C77G12 against SARS-CoV-2 WT, Delta and Omicron variant S protein-pseudotyped viruses, as evaluated in 293T-ACE2 cells.

To clarify the experimental context:

Initially, we determined the WT S<sup>E-FIC</sup>-76E1 structure using 11,263 selected micrograph from dataset1 (WT-ACE2-76E1, 30min), in which no postfusion particles were observed. Under identical conditions, we attempted to resolve the XBB.1.5 S<sup>E-FIC</sup>-76E1 structure using dataset3 (XBB.1.5-ACE2-76E1, 30min, total 2,283 good micrographs). However, we were unable to obtain a high resolution XBB.1.5 S<sup>E-FIC</sup>-76E1 structure suitable for structural analysis; and in contrast to WT, the postfusion particles were observed. Thus, to eliminate the possibility that the observed postfusion particles was solely due to the intrinsic instability of XBB.1.5 S, we collected an additional dataset2 (XBB.1.5-ACE2-76E1, 10min, total 7,267 micrographs).

We analyzed a subset of micrographs matched in number to ensure comparability. The increase in postfusion particles under these conditions supports the observation that 76E1 cannot fully block the conformational transition of the XBB.1.5 S, consistent with its reduced neutralization efficacy against this variant.

(3) Response to “What is the molecular rational for (Lines 204-210)?”  
(Despite similarities in 2D classification and 3D reconstruction between WT and

XBB.1.5 L-FIC-76E1, attempts to obtain a high-resolution structure of XBB.1.5 S<sup>E-FIC</sup>-76E1 were unsuccessful, indicating its heterogeneity. 76E1 binding requires S1-ACE2 to either detach or extend outward from the S-trimer three-fold axis, with the latter increasing E-FIC particle heterogeneity. The reduction in E-FIC and increase in the postfusion state suggest that XBB.1.5 binds 76E1 less efficiently than the WT, likely due to reduced epitope exposure. (Lines 204-210))

We thank the reviewer for pointing out these incorrect descriptions again. We have removed the descriptions in the revised manuscript. Instead, we resolved the XBB.1.5 S<sup>E-FIC</sup>-76E1 structure at modest resolution. Comparative structural analysis based on the newly determined XBB.1.5 S<sup>E-FIC</sup>-76E1 structure has been included into the *Cryo-EM structures of XBB.1.5 intermediate S recognized by 76E1-Fab* section of the revised manuscript, as described previously.

**11, Why would 76E1 binding to XBB.1.5 spike require detachment of S1-ACE2 or further outward extension than WT? What evidence supports this statement? Is there an Extended Data Figure describing data collection and processing of this XBB.1.5 cryoEM dataset?**

**Response:**

Thanks for the reviewer's valuable suggestion. As mentioned above, we determined the XBB.1.5 S<sup>E-FIC</sup>-76E1 structure using optimized sample preparation methods. The details have been included into the *Methods* section, and data collection and processing of this XBB.1.5 cryoEM dataset have been summarized in **Extended Data Fig. 5 and Supplementary Table 2.**

Comparison to the WT S<sup>E-FIC</sup>-76E1, we found that in XBB.1.5 S<sup>E-FIC</sup>-76E1, the S2-top region containing the S2'-helix epitope is rotated counterclockwise by approximately 6.4° around the 3-fold axis and the S2'-helix moves downward by ~1.2 Å; however, the S1-ACE2s move upward by ~2.6 Å, which leads to a smaller and inappropriate space for 76E1 binding and creates steric hindrance against the 76E1 antibody (**Fig. 2d-i and Extended Data Fig. 7**). Consequently, the S1-ACE2s undergo

a counterclockwise rotation of  $\sim 3.8^\circ$  with ACE2s rotating outward  $\sim 2.2^\circ$ , leading to an additional increase in the minimum distance between ACE2 and the 3-fold axis from 36.9 Å to 41.2 Å to accommodate 76E1, which entails surmounting greater energy barriers. The statement “detachment of S1-ACE2” has been removed, and the related descriptions are included into the *Cryo-EM structures of XBB.1.5 intermediate S recognized by 76E1-Fab* section of the revised manuscript.

**12, Lines 226-228:** Are the authors referring to differences of relative orientation between constant and variable domains (i.e., elbow angle)? Could it be that the 3 Fabs compared crystallized with distinct packing, hence the different elbow angles observed, but that they would be able to flex around it to bind to EIC (as the elbow angle is notoriously flexible for virtually all mAbs)?

**Response:**

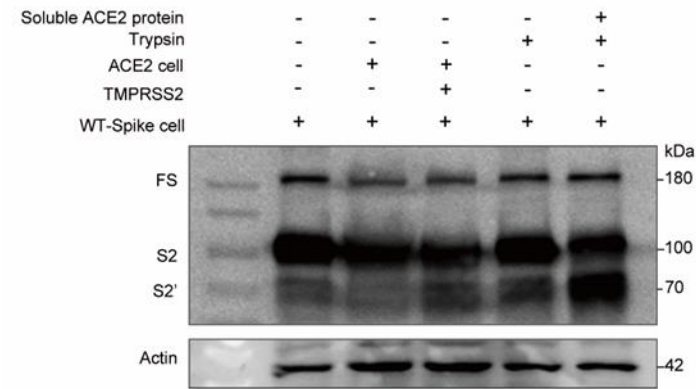
We sincerely appreciate the reviewer’s insightful comments, which have been pivotal in refining our manuscript. We agree with your viewpoint. It was indeed our oversight in not thoroughly considering that particular section. We have removed related data and descriptions in the revised manuscript.

**13, Lines 241-245:** Have the authors tested cleavage with TMPRSS2 found on target cells (trans) as opposed to the spike-producing cells (cis)?

**Response:**

We have ever tried both trans and cis methods to test the cleavage with TMPRSS2. For the trans experiment, 293T cells stably expressing ACE2 were transiently transfected with TMPRSS2 plasmids for 24 h (293T-ACE2-TMPRSS2). Subsequently, 293T-ACE2-TMPRSS2 cells were incubated with 293T cells stably expressing spike (293T-spike) for cleavage for 2 h. Soluble trypsin incubated with 293T-spike cells was included as the positive control. However, we failed to observe the TMPRSS2 cleavage at the S2’ site in this experiment, while the soluble trypsin can effectively cleave at the S2’ site, showing as significant S2’ segment (**Fig. R3**). The failure in the S2’ site cleavage of this trans method might be attributed to the reason that the over expressed

TMPRSS2 may also cleave ACE2 protein during long time (24h) TMPRSS2 transfection and cell culture, thus inactive ACE2 protein and unable to prime spike for expose S2' site. Therefore, we ultimately chose the cis method to test the cleavage with TMPRSS2, as shown in **Fig. 3a**.



**Fig. R3.** Trans cleavage of cell-expressed WT spike protein at the S2' site by TMPRSS2, with or without ACE2-Fc, assessed by Western blot.

14, Lines 252-262: TMPRSS2 is indeed known to be expressed in Caco2 cells but cathepsins are too (hence the inhibitory effect of camostat and E64d in Fig3e for BF.7 and XBB.1.5)

**Response:**

Thanks for the reviewer's insightful comments. TMPRSS2 and cathepsins are indeed known to be expressed in Caco-2 cells. The camostat/E64d inhibition assays in **Fig. 3e** serve to validate the functionality of our experimental system. These data directly support findings in **Fig. 3f-g**, demonstrating that viral entry route determines the neutralizing efficacy of the 76E1 antibody. Critically, the increased CTSL-mediated entry pathway utilization of Omicron variants contributes to 76E1's diminished efficacy, a constraint effectively countered by reducing antibody molecular size. We concur with the reviewer that the cited expression data are well-established in the literature. Consequently, we have removed this dataset from **Fig. 3**, but incorporated appropriate references<sup>3,4</sup> (PMID: 32142651 and PMID: 34159616) in the *Main text*. (Lines 266-267)

15, The authors seem to imply that a switch from a TMPRSS2-dependent to a CTSL-dependent entry pathway occurred to evade S2' helix directed antibodies. However, all reports have indicated that these types of antibodies are both rare and weakly neutralizing and it is therefore unlikely that they would be able to exert an appreciable selective pressure that would have led to this outcome.

**Response:**

We appreciate the reviewer's valuable suggestion and sincerely apologize for the misunderstanding caused by our unclear description, which led to misinterpretation by the reviewer. We agree with the reviewer's opinion that such antibodies are rare in the human body, and it is therefore unlikely for them to exert selective pressure on the virus. To clarify, our intention was to convey that the Omicron variants have undergone changes in its entry pathway, which subsequently resulted in a reduced neutralizing activity of these antibodies, rather than the antibodies driving the alteration in the virus's entry pathway.

To ensure manuscript accuracy, we have revised the text to attribute the diminished neutralization efficacy of 76E1 to altered entry routes of Omicron, rather than depicting immune evasion as an active viral process.

16, How do the authors reconcile the proposed reduced/loss of TMPRSS2 dependence for Omicron variants (observed by several other groups) with contradictory results in organoid models (PMID: 37555660) and in vivo (PMC9858505)?

**Response:**

We sincerely appreciate the reviewer's valuable insights. Although Omicron variants have shifted towards a CTSL-dependent endosomal pathway, they still exhibit sensitivity to TMPRSS2, rather than indicating a complete loss of TMPRSS2 utilization<sup>2,9,10</sup>. The differing TMPRSS2 dependency observed in cell lines versus organoid and in vivo models may be attributed to variations in the expression levels of CTSL and TMPRSS2. TMPRSS2 is widely expressed on the luminal surface of respiratory epithelial cells, which are directly exposed to the airway

microenvironment<sup>11,12</sup>. In contrast, CTSL is moderately expressed in ciliated cells and goblet cells of the upper respiratory tract, especially the nasal cavity. For the lower respiratory regions, CTSL is relatively enriched in AT2 cells. However, AT2 cells occupy only ~7% of the alveolar surface area<sup>12-14</sup>. Therefore, TMPRSS2 is typically highly expressed throughout the respiratory tract, especially in the lower airways. Conversely, CTSL expression is generally low overall but elevated in the nasal cavity.

The airway organoid models used in the referenced study (PMC9858505) were derived from airway stem cells of the lung, primarily recapitulating the microenvironment of the conducting airways (trachea, bronchi, and bronchioles)<sup>15</sup>. In these regions, TMPRSS2 expression is elevated, whereas CTSL expression remains low, a pattern similarly observed in intestinal organoids<sup>16</sup>. Consequently, inhibitor sensitivity assays conducted on these types of models may overestimate the efficacy of TMPRSS2 inhibitors while underestimating the effects of CTSL inhibitors<sup>15</sup>. Another study indicates that Omicron variants efficiently infect nasal organoids characterized by high CTSL expression<sup>17</sup>. Thus, the observed discrepancies across different organoid models and in vivo models are possibly attributed to tissue-specific expression patterns of these proteases.

**17, Extended Data Fig7a: Why are the binding responses (OD) so small in these ELISAs?**

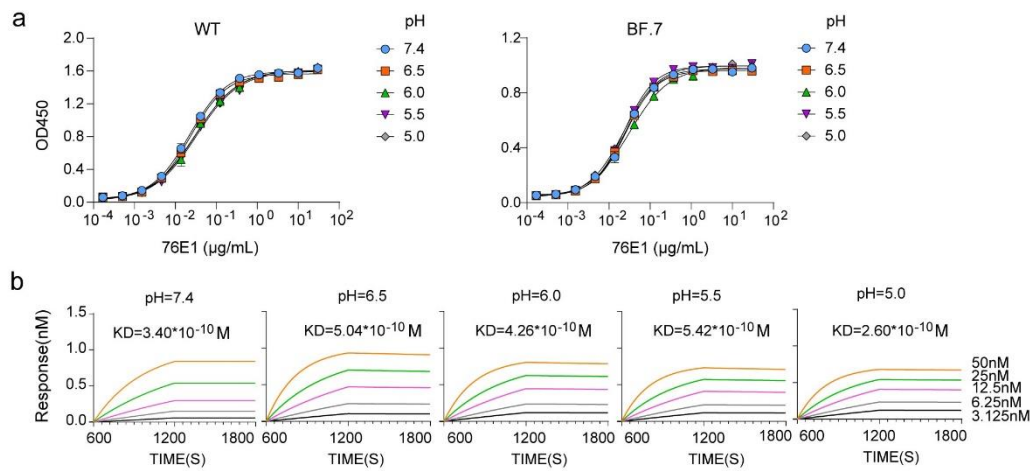
**Response:**

The small binding signals observed in these ELISA assays are likely attributable to two methodological parameters: (1) relatively low antigen coating density, and (2) abbreviated substrate reactive time. Although the maximum OD<sub>450</sub> values (1.0-1.6) in our study are comparatively not high, they remain within the documented range for valid immunoassays, as evidenced by published studies reporting OD<sub>450</sub> maxima between 0.8-3.5<sup>18,19</sup>. Moreover, the dose-response relationships demonstrated across experiments provide compelling validation of data reliability.

**18, Why using BA.1 S instead of BF.7 in a-b?**

**Response:**

We sincerely apologize for the inconsistency in Omicron S protein variants used in our initial assays. To ensure methodological rigor, we repeated these experiments using BF.7 S protein. Critically, the results recapitulate our original findings: the reduced pH conditions (pH: 5.0-7.4) are not a major factor to alter the binding affinity of 76E1 antibody towards Omicron spike proteins (**Extended Data Fig. 8a-b**). We have revised this description in *Endosomal acidity plays a limited role in Omicron's evasion from 76E1* and *Methods* sections of the revised manuscript. (Lines 276-289)



**Extended Data Fig. 8a-b a**, ELISA-based binding curves of 76E1-FL antibody to SECD proteins of WT SARS-CoV-2 and Omicron BF.7 under different low pH conditions. pH 7.4 was included as the control.  $n=3$ . Data are shown as mean  $\pm$  SEM. **b**, Binding affinity of 76E1-FL antibody to Omicron B F.7 SECD protein as measured by BLI under different low pH conditions. pH 7.4 was included as the control.

**19, Lines 304-307: These two sentences seem contradictory as CTSL entry will always be the route used in 293T-ACE2 cells**

**Response:**

We sincerely apologize for the lack of clarity in our initial description, which led to potential ambiguity. In the revised manuscript, we focused on comparing the entry pathways and antibody neutralization between: (i) WT and its H655Y-mutant viruses, and (ii) BF.7 and its Y655H-reverted variant, to determine the functional impact of the

H655Y mutation in 293T-ACE2 and Caco-2 cells. 293T-ACE2 cells exclusively support viral entry via the endosomal pathway, whereas Caco-2 cells permit both cell surface and endosomal entry routes, with the cell surface pathway being dominant. We have reorganized the figure, which is **Fig. 4e** in the revised manuscript. The modified descriptions are as follows: “*Comparing of WT and its H655Y-mutant pseudovirus confirmed that H655Y mutation increases viral reliance on the endosomal pathway while reducing dependence on the plasma membrane pathway, manifested by increased sensitivity to the CTSL inhibitor E64d and decreased sensitivity to the TMPRSS2 inhibitor camostat. The BF.7-Y655H reversion mutant pseudovirus further validated this phenotype.*” (**Lines 317-323**)

**20, Lines 319-321:** Have the authors imaged the ACE2/S+H655Y to confirm that S1 is not moving as much as in the structures they obtained here and in their recent Cell paper?

**Response:**

Thanks for the reviewer’s valuable suggestion. In the revised manuscript, we resolved the S<sup>E-FIC</sup>-76E1 structure of XBB.1.5, containing the H655Y mutation at modest resolution. Structural comparisons of the apo D614G S<sup>E-FIC</sup>, WT S<sup>E-FIC</sup>-76E1 and XBB.1.5 S<sup>E-FIC</sup>-76E1 reveal that the S1-ACE2s in apo E-FIC need to move downward along the axis and rotate outward away from the axis to create adaptive space to accommodate 76E1 (**Extended Data Fig. 7a**). However, for the XBB.1.5 S<sup>E-FIC</sup>-76E1 structure, the S1-ACE2s move downward less than the WT, which implies that the gap between the S2’-helix and the ACE2 for XBB.1.5 is smaller than that of WT before the S1-ACE2s rotating outward. Therefore, the S1-ACE2s need to surmount greater energy barriers to move outward to accommodate 76E1 for XBB.1.5. Our present data supports the interpretation of changes in S2’-helix accessibility. We rephrased this statement in the revised manuscript.

“*This bond may make S1-ACE2 less likely to move downward along the 3-fold axis, shown as **Extended Data Fig. 7a**, potentially explaining why the H655Y mutation increases the steric hindrance at the S2’-helix epitope, thereby rendering it inaccessible*

1064 *to large-sized antibody, which requires a larger accommodation space.” (Lines 340-*  
1065 **343)**

1066 **21**, Lines 323-324: The data presented in Fig4f indicate that all pseudoviruses tested  
1067 depend on endosomal entry only in (293T-ACE2 cells) and are affected by inhibitors of  
1068 both plasma membrane (camostat) and endosomal (E64d) entry. This sentence therefore  
1069 seems to be an overstatement.

1070 **Response:**

1071 Thanks for your valuable suggestions. The text has been revised as follows: “*These*  
1072 *findings indicate that the enhanced utilization of CTSL-dependent entry pathway by*  
1073 *H655Y mutation enables Omicron variants to escape 76E1.*”. Here we emphasize that  
1074 the H655Y mutation facilitates a switch preference in viral entry pathways, avoiding  
1075 overstatement that it directly causes a complete alteration of these pathways. (Lines  
1076 **330-332)**

1077 **22**, What evidence do the authors have to support the lower opening of S1 for XBB vs  
1078 WT?

1079 **Response:**

1080 Thanks for pointing out this issue. As mentioned before, the S1-ACE2s in apo E-  
1081 FIC must move downward along the axis and rotate outward away from the axis to  
1082 create adaptive space to accommodate 76E1 (**Extended Data Fig. 7a**). However, for  
1083 the XBB.1.5 S<sup>E-FIC</sup>-76E1 structure, the S1-ACE2s move downward less than that of the  
1084 WT, which implies that the gap between the S2'-helix and the ACE2 for XBB.1.5 is  
1085 smaller than that of WT before the S1-ACE2s rotating outward for 76E1 binding.  
1086 Therefore, the S1-ACE2s need to surmount greater energy barriers to open outward to  
1087 accommodate 76E1 for XBB.1.5. The results support the larger opening of S1-ACE2s  
1088 for XBB.1.5 compared to WT. We have corrected the related statements in the revised  
1089 manuscript.

1090 **23**, Lines 377-379: The accumulation of E-FIC should not be an issue for 76E1 binding  
1091 as it is a structure that was obtained by the authors. This does not make sense and is in

discrepancy with the data shown in Fig2a: 25.4% (WT) vs 15.3-20.7% (XBB.1.5) for E-FIC. Refolding to postfusion is the issue as 76E1 cannot bind to this conformation.

**Response:**

Thanks for the suggestion. We agree with the reviewer's comment, and have removed the related descriptions in the revised manuscript.

**24,** Based on the file names and on the appearance of the maps, it appears that the authors provided deep learning enhanced reconstructions. While the use of such maps can help model building, unmodified maps should be provided and deposited to the EMDB. This reviewer encourages the author to do the same and update the maps deposited for their recently published Cell paper.

**Response:**

Thanks for the suggestion. All unmodified maps have been provided and deposited to the EMDB.

**Referee #4 (Remarks to the Author):**

The continued emergence of coronaviruses in the human population demonstrates the importance of developing vaccines and therapeutics that confer broad protection. The spike protein, responsible for receptor binding and cell fusion, has been the focus of vaccines and monoclonal antibody therapies. However, the antigenic evolution of the spike has had a significant impact on vaccine efficacy and has rendered all clinical-stage monoclonal antibodies (mAbs) ineffective. Identifying and characterizing conserved regions in the spike is crucial for developing more robust countermeasures.

One of the most promising targets within the spike is the S2'-helix, a pan-coronavirus epitope buried in the spike, which becomes transiently exposed upon ACE2 binding. While the binding of these antibodies has been characterized in the context of isolated peptides, the molecular basis for their inhibitory mechanism of action in the

context of spike refolding remained unknown – in part due to the transient nature of spike fusion intermediates. Moreover, mechanistic studies into potential immune evasion of these important antibodies by SARS-CoV-2 variants were lacking.

Through structural and functional studies, Bao et al. reveal that Omicron variants evade S2'-helix-targeting antibodies by reducing epitope exposure, speeding up spike refolding, and shifting to a Cathepsin L-dependent entry pathway, with the H655Y mutation playing a crucial role. Through antibody minimization experiments, the authors demonstrate that protein size affects its access to the S2'-helix. Extensive cryoEM analysis shows that these antibodies stabilize both early and late intermediates of the SARS-CoV-2 spike, but they capture fewer early intermediates of the Omicron spike due to reduced epitope exposure and faster refolding. Notably, minimizing antibody size reversed these negative effects and significantly boosted neutralizing activity against Omicron variants and other human coronaviruses, including SARS-CoV-1 and HCoV-229E.

This impressive work provides important new information about this crucial epitope and offers tangible strategies for designing pan-coronavirus biologics by targeting the S2'-helix, paving the way for more efficient and broad therapeutic strategies. As such, I believe this work would fit within the remit of Nature once several issues have been addressed.

**Response:**

Thanks for the positive comments on our manuscript.

**Main concern**

1, In Figure 1A [2A] and Extended Data Figure 3, the authors quantify different conformation states of the spike protein. Based on inspection of the 2D and 3D class averages, I mostly agree with the authors' interpretation. However, the clear anisotropy for certain conformations, such as the XBB.1.5 E-FIC and L-FIC, leaves some room for interpretation. For example, how do the authors know that the L-FIC is a true late intermediate and not the product of damaged particles? Conceivably, absorption to the

air-water interface could result in such classes. I think it would be wise to include such a short caveat in the main or supplementary text to cover this possibility, perhaps in lines 183-185, where the authors suggest the lack of density is due to “a dynamic characteristic.”

**Response:**

Thanks for the reviewer’s valuable suggestion.

[REDACTED]

(2) For the XBB.1.5 S<sup>E-FIC</sup>-76E1, we further resolved its structure by optimized methods. The details have been included into the *Methods* section, and data collection and processing of this XBB.1.5 cryoEM dataset have been summarized in **Extended Data Fig. 5 and Supplementary Table 2**. Comparison to the WT S<sup>E-FIC</sup>-76E1, we found that in XBB.1.5 S<sup>E-FIC</sup>-76E1, the S2-top region containing the S2’-helix epitope

is rotated counterclockwise by approximately  $6.4^\circ$  around the 3-fold axis and the S2'-helix moves downward by  $\sim 1.2$  Å; however, the S1-ACE2s move upward by  $\sim 2.6$  Å, which leads to a smaller and inappropriate space for 76E1 binding and creates steric hindrance against the 76E1 antibody (**Fig. 2d-i and Extended Data Fig. 7**). Consequently, the S1-ACE2s undergo a counterclockwise rotation of  $\sim 3.8^\circ$  with ACE2s rotating outward  $\sim 2.2^\circ$ , leading to an additional increase in the minimum distance between ACE2 and the 3-fold axis from 36.9 Å to 41.2 Å to accommodate 76E1, which entails surmounting greater energy barriers. The related descriptions are included into the *Cryo-EM structures of XBB.1.5 intermediate S recognized by 76E1-Fab* section of the revised manuscript.

2, Moreover, given the heterogeneity and importance of these data sets, I would strongly advise that the authors deposit the raw data from Figure 1A [2A] and Extended Data Figure 3 to EMPIAR. The reason for this is that unlike the larger 11,263 data set, the community has no way to assess the product of the processing, which in this case is 2D and 3D class averages.

**Response:**

We appreciate the reviewer's suggestion on data deposition. We understand this request because we did not determine the structure of XBB.1.5 S<sup>E-FIC</sup>-76E1 in the previous version of our manuscript.

In the revised manuscript, we have improved the sample preparation procedures, recollected cryo-EM data, and successfully resolved the XBB.1.5 S<sup>E-FIC</sup>-76E1 structure to an overall resolution of 5.65 Å. Importantly, local refinement of the S2-top region and the 76E1 epitope improved the resolution to 4.14 Å, which is sufficient for detailed structural analysis.

As a result, we removed the previous comparison based on 2D class averages (Fig. 2a), which were less accurate and less informative. Instead, we now present a comparison based on the 3D reconstructions in the revised manuscript. Both unsharpened and sharpened density maps have been deposited in the EMDB, providing

adequate support for the structural interpretations presented in the study.

That said, we are happy to deposit the raw cryo-EM data to EMPIAR should the reviewer or editor feel that this would further strengthen the transparency and reproducibility of our results.

#### Minor concerns

3, Related to my main concern, in lines 204-207, the authors state: “Despite similarities in 2D classification and 3D reconstruction between WT and XBB.1.5 L-FIC-76E1, attempts to obtain a high-resolution structure of XBB.1.5-E-FIC-76E1 were unsuccessful, indicating its heterogeneity.” I presume the authors intend to refer to only E-FIC in this passage; otherwise, it does not make sense. Moreover, I would say that the primary reason for the inability to obtain a structure of E-FIC is due to a lack of side views and not its heterogeneity.

#### Response:

Thanks for pointing this out. We agree with the reviewer’s comment. Indeed, orientation preference was a main reason that we did not determine a high-resolution structure of XBB.1.5 S<sup>E-FIC</sup>-76E1. To solve this problem, we optimized the grid preparation condition and data processing and obtained a higher resolution structure of XBB.1.5 S<sup>E-FIC</sup>-76E1, which allows us to do model building and structural analysis. Thus, we deleted this statement in the revised manuscript.

4, In lines 169-174, the authors write the description of the epitope-paratope interactions in a manner that suggests this is all readily interpretable from the cryo-EM maps. Given the modest resolution, and the fact the maps have used density modifying software, which I am not generally in favour of, it is a bit misleading. Moreover, this is not necessary as this information is already available from the crystal structure, used as a starting model and mentioned in lines 180-181. The strength of the cryo-EM data in the present study is that it provides new insights into the quaternary structure of the complex at modest resolution. The description of the atomic details can be shortened and toned down.

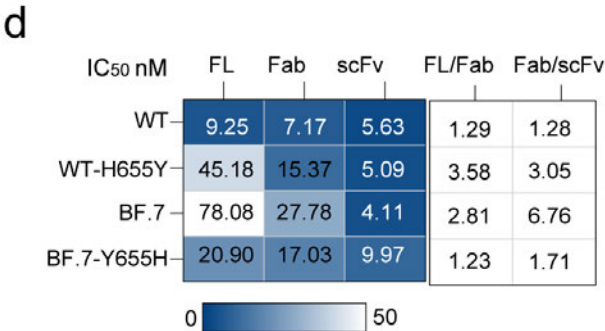
**Response:**

Thanks for the valuable suggestion. As suggested, we have deleted the descriptions of the atomic details and rephrased this statement to “*The cryo-EM and crystal structures of S2'-helix-76E1 are very similar, with some subtle differences in the binding interactions between S2'-helix and 76E1.*” in the revised manuscript. (Lines 172-174)

5, In lines 300-303, the authors state: “Similar results were found in the neutralization assay against H655Y pseudovirus in 293T-ACE2 cells, with scFv demonstrating a more significant enhancement of neutralizing activity (Fig. 4e).” To me, the phrasing of this sentence implies that they are comparing the FL to scFv enhancement in H655Y to something else, presumably the WT. Should the authors not then point to another figure to demonstrate this, such as Figure 1F?

**Response:**

We appreciate the reviewer’s valuable suggestion. In response, we have repeated this experiment and included the WT pseudovirus as the control. As the figures have been reorganized, this figure is shown in **Fig. 4d**. The relevant text has been revised to explicitly illustrate this comparison in the updated manuscript. (Lines 309-314)

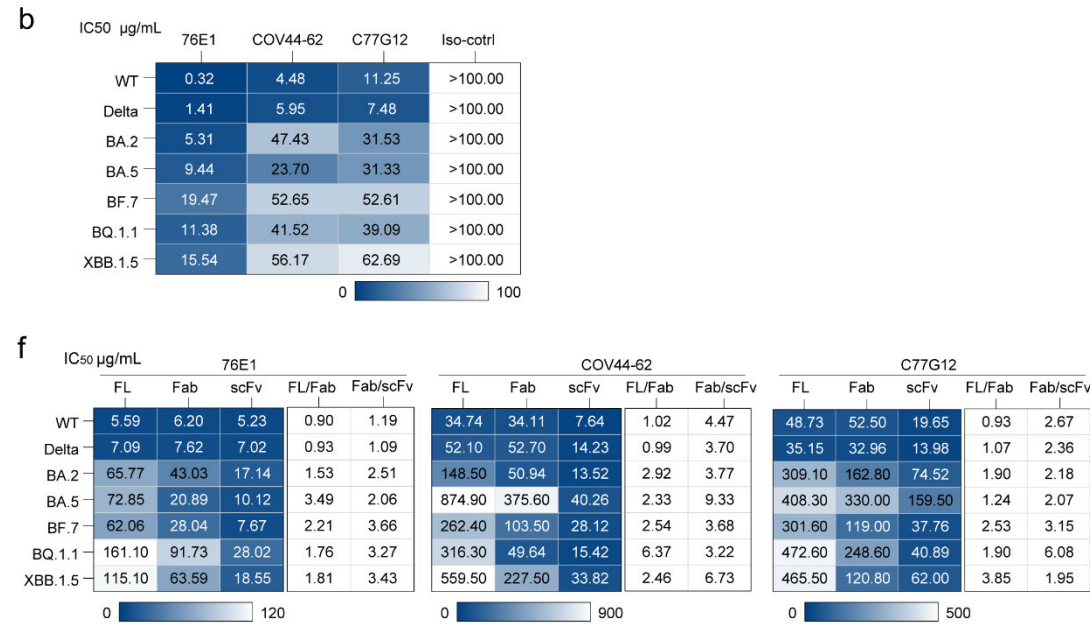


**Fig. 4d**, Neutralizing IC50 values of 76E1-FL, Fab and scFv at equal molar against indicated pseudoviruses in 293T-ACE2 cells. Data are shown as mean ± SEM.

6, It is a little confusing that the authors choose orange to indicate high potency in Figure 1B and reduced potency in Figure 1F.

**Response:**

We appreciate the reviewer’s valuable suggestion. In both **Fig. 1b** and **1f**, we have revised the figures to show high neutralizing activity in dark blue and low neutralizing activity in light colors.



**Fig. 1b**, Neutralizing IC<sub>50</sub> values of full-length 76E1, COV44-62 and C77G12 against SARS-CoV-2 WT, Delta and Omicron pseudoviruses, as evaluated in 293T-ACE2 cells.

**Fig. 1f**, Neutralizing IC<sub>50</sub> values of 76E1, COV44-62, and C77G12 (equal molar concentrations; varying molecular sizes) against SARS-CoV-2 WT, Delta, and Omicron pseudoviruses in 293T-ACE2 cells.

7, Remove mention of “high-resolution” in line 368 as it is misleading and not necessary.

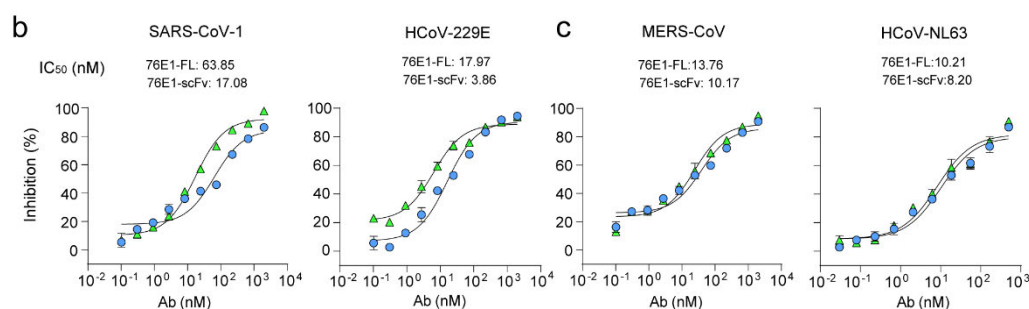
**Response:**

Thanks. we have removed “high-resolution”.

8, In line 341, the authors refer to MERS-CoV and HCoV-NL63 in panel 5C, but this is not correct. I think the MERS-CoV and HCoV-229E panels need switching for this to make sense.

**Response:**

Thanks for pointing this out. We have already switched the positions of MERS-CoV and HCoV-229E panels to make sense, as shown in **Fig. 5b-c**.



**Fig. 5b-c.** **b**, Neutralization of *betacoronaviruses* pseudoviruses (SARS-CoV-1, HCoV-229E) by 76E1-FL and scFv in 293T-ACE2 cells. **c**, Neutralization of *alphacoronaviruses* (MERS-CoV, HCoV-NL63), with MERS-CoV in HuH-7 cells, others in 293T-ACE2 cells.  $n = 3$ . Data are shown as mean  $\pm$  SEM.

9, There appear to be several errors in the reference DOIs.

**Response:**

Thanks for pointing this out. We have revised the references.

10, In the cryo-EM sample preparation and data acquisition methods, small adjustments could be made to make the methods a more accurate. Firstly, if 1.19 Å and 0.932 Å 'pixel sizes were calibrated using a crystal structure, replace “physical pixel size” with “calibrated pixel size.” If these pixel sizes were read from EPU, then you should simply say “a nominal magnification of 105,000x with a pixel size of 1.19 Å.” Physical is often used to refer to the physical pixel size on a detector, which is 14 μm. Secondly, I am assuming that the EER movie was fractionated into a number of dose fractions resulting in  $\sim 1 \text{ e}/\text{Å}^2$ ? If so, it should read “EER movie stack consisting of 1,737 EER frames was dose-fractionated to 50 fractions” or something similar.

**Response:**

Thanks for pointing this out. We have changed this statement “physical pixel size” to “calibrated pixel size” and updated “each EER movie stack was dose-fractionated to 1,737 frames.” to “each EER movie stack consisting of 1,737 frames was fractionated to 43 subgroups.” in the revised manuscript. (Lines 654-663)

11, Did the authors really collect the exact same number of movies for the WT and

XBB.1.5 data shown in Figure 2A? I can imagine this was a conscious choice to ensure equal sample sizes or simply a typo.

**Response:**

Thanks for pointing this out. Indeed, we collected different numbers of movies for each dataset. To enable a fair comparison between different samples, we used the same number of movies across datasets to ensure equivalent sample sizes. Additional details have been included in the related sections of the *Methods* section in the revised manuscript.

**12,** A general comment is that the writing could be improved. A non-exhaustive list of issues is included below:

Line 24: “this pathogen” implies a single agent.

Line 55: “of noting” should be “of note.”

Line 73: “variants has” should be “variants have.”

Line 97: You can remove the “of” after “sterically hindering.”

Line 116: “FACS” is written as if it is an abbreviation for flow cytometry.

Line 662: “Captured” can be replaced with “collected.”

Extended data line 98: Erroneous punctuation after “and.”

**Response:**

Thanks for pointing out these issues. We have corrected the errors mentioned by the reviewer. In addition, we have carefully reviewed the entire manuscript and revised typos, errors, and inconsistent terms.

**13,** Remarks on code availability: Cryo-EM maps and models have been deposited the EMDB and PDB, but I am recommending that raw cryoEM data for two of the data sets should be deposited to EMPIAR.

**Response:**

We sincerely appreciate the reviewer's suggestion regarding the deposition of raw cryoEM data to EMPIAR. As described above, we have now successfully resolved the structure of XBB.1.5 S<sup>E-FIC</sup>-76E1 at a resolution suitable for structural analysis. In this case, the deposited cryo-EM maps and atomic models in EMDB and PDB should be sufficient to support the conclusions presented in the manuscript.

That said, we are happy to deposit the raw cryo-EM data to EMPIAR should the reviewer or editor feel that this would further strengthen the transparency and reproducibility of our results.

## References

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1379 *of Sciences of the United States of America* **120**, e2300376120,

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1381   18       Zhang, C. *et al.* Development and structural basis of a two-MAb cocktail for

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1384   19       Shi, W. *et al.* Vaccine-elicited murine antibody WS6 neutralizes diverse beta-

1385           coronaviruses by recognizing a helical stem supersite of vulnerability.

1386           *Structure (London, England : 1993)* **30**, 1233-1244.e1237,

1387           doi:10.1016/j.str.2022.06.004 (2022).

1388

## **Response to referees' comments**

### **Referee #1 (Remarks to the Author):**

The authors examine a cryptic, conserved epitope that becomes exposed during the conformational changes of fusion.

The most exciting part of the paper was indeed, the fusion intermediate structure. An apo form of this structure was published by some of the authors in Cell. What is different in this paper from the previous is that the intermediate structure is bound by scFv of the antibody of interest. There is also analysis of how Omicron evades such antibodies by some conformational masking (as opposed to mutating amino acid contacts), and by a relative change in protease preference which undoes the ability of the antibody to neutralize by blocking the previously preferred protease.

Lessons applicable to other fields include revelation of antibody recognition of an epitope exposed in a fusion intermediate, exploration of mechanism of antibody escape that are more complex than change in amino acid contact, the finding that smaller versions of the antibody can access cryptic epitopes, and highlighting this particular helix as a potential small molecule target.

Although the aspect I was most excited about - the pre-fusion. intermediate structure, does seem to have been revealed before in an unbound way, the antibody complex and analysis of how antibody is evaded are useful and thoughtful contributions to the field.

I am focusing here on the big picture and not on data specifics.

### **Response:**

We sincerely appreciate the positive comments on our revised manuscript.

### **Referee #2 (Remarks to the Author):**

I appreciate their extensive effort in addressing the prior comments and improving the quality and clarity of the study. Overall, the authors have appropriately addressed all of my previous concerns and substantially strengthened the manuscript. The inclusion of new structural data, expanded analyses of antibody–spike interactions, and clarification of the “undefined class” add rigor and transparency to the work. Regarding the issue of novelty, I do note some partial overlap with the group’s previously published study. However, this revised manuscript appears to include additional and higher-resolution structural insights, particularly concerning the S2'-helix conformation and mechanisms of antibody evasion by Omicron subvariants. In my view, the revised manuscript is much improved and now provides valuable mechanistic insights that will interest the virology and structural biology communities.

### **Response:**

We sincerely appreciate the positive comments on our revised manuscript.

Referee #2 (Remarks on code availability):

Appropriate

**Response:**

We sincerely appreciate the positive comments on the code availability of our revised manuscript.

Referee #3 (Remarks to the Author):

This reviewer notes that the authors worked diligently to address the extensive reviewers' comments.

This reviewer also provided additional comments below to help the authors strengthen the manuscript, which reports many important findings. Although the relevance of CTSL-mediated entry remains uncertain—given that TMPRSS2-mediated entry appears to be the predominant pathway based on in vivo, ex vivo, and in vitro data—it is nevertheless a thought-provoking study that will be of broad interest.

**Response:**

We are grateful for your comments, which have significantly strengthened our manuscript. In response, we have conducted additional experiments, incorporated relevant literature, and revised the text accordingly. Our point-by-point responses are detailed below.

1. A growing body of in vitro and in vivo data indicate that Omicron variants still use TMPRSS2 and suggest that CTSL utilization is an artifact of some assays (e.g. PMID: 36243815, <https://www.biorxiv.org/content/10.1101/2025.07.07.663433v1>, PMID: 37769654, PMID: 37555660)

**Response:**

We sincerely thank the reviewer for raising this important point and for directing our attention to the recent literature on Omicron entry pathways. We fully agree that Omicron lineages retain the ability to use TMPRSS2 and that TMPRSS2-mediated entry remains a critical and physiologically important route, particularly in primary human respiratory tissues and in vivo models. Our study did not intend to claim that Omicron has lost TMPRSS2 usage, but rather to describe a context-dependent change in protease preference and its functional impact on the accessibility of the S2'-helix epitope.

We have carefully examined the papers cited by the reviewer. Below is a point-by-point analysis: **(1)** PMID: 36243815 and the preprint (bioRxiv 2025.07.07.663433) clearly establish TMPRSS2 as a major protease for Omicron entry. However, these studies did not directly assess the potential contribution of the endosomal pathway, leaving this question open. We note that PMID: 36243815 reports in its murine model that Omicron variants use TMPRSS2 less efficiently than earlier variants such as Beta and Gamma, suggesting the possible involvement of other entry routes for Omicron. **(2)** PMID: 37769654 concluded that endosomal entry is not critical in Omicron because the cathepsin B inhibitor CA-074-Me showed limited inhibitory effect. However, CA-074-Me is specific to cathepsin B (CTSB) and does not inhibit cathepsin L (CTSL). Previous work (e.g., PMID: 33465165) indicates that when TMPRSS2 expression is low, the CTSL inhibitor MDL28170 has a substantially greater impact on viral entry than CA-074-Me. For instance, 1  $\mu$ M CA-074-

Me had no effect, while 1  $\mu$ M MDL28170 inhibited ~60% of SARS-CoV-2 entry. Thus, broad-spectrum cysteine protease inhibitors like E64d may be more appropriate for assessing the overall contribution of the endosomal pathway. **(3)** For PMID: 37555660, which employed CTSL/B knockout intestinal organoids, the evidence for complete knockout efficiency was not provided, introducing uncertainty in interpreting the reported results.

In addition, several other studies using more physiologically relevant models support the role of the endosomal pathway in Omicron entry: **(1)** PMID: 39601592 showed that the endosomal inhibitor E64d significantly blocks Omicron infection in human respiratory organoids, with a higher inhibitory effect against Omicron than against the ancestral strain, indicating enhanced reliance on this pathway. **(2)** PMID: 39753670, using human primary respiratory epithelial cells, reported similar findings.

In summary, evidence from human primary tissues, organoids, and in vivo models supports a model in which TMPRSS2-mediated fusion remains a key pathway for Omicron, while its efficiency relative to earlier variants is reduced, accompanied by a concurrently enhanced utilization of the endosomal pathway. We agree with the reviewer that CTSL dependency can be overestimated in certain artificial systems if not carefully interpreted. Our cleavage and neutralization experiments were conducted in defined cell systems (293T-ACE2, CTSL-dominant; Caco-2, TMPRSS2-high) to mechanistically dissect how pathway bias affects S2'-helix accessibility and antibody efficacy. Our conclusions pertain to these specific experimental contexts and the functional consequences of a changed protease preference, rather than claiming a universal entry hierarchy across all tissues. We acknowledge that the heterogeneous expression of TMPRSS2 and cathepsins across tissues leads to variation in entry routes, which would consequently affect the in vivo neutralization profile of antibodies like 76E1 against Omicron in a tissue-dependent manner.

#### **Revisions made to the manuscript:**

In response to the reviewer's concern, we have revised the text to more accurately reflect the above understanding:

##### **(1) Explicit acknowledgement of TMPRSS2 importance and softened wording:**

We now explicitly state that TMPRSS2 is critically important for infection in vivo and in primary human respiratory tissues for Omicron, as follows,

Abstract: Change "...combined with a shift to Cathepsin L-dependent entry pathway..." to *"...combined with an enhanced preference for cathepsin-mediated endosomal entry..."*

Introduction: Replace "Emerging data indicate that the Omicron variants have switched entry route preference..." with *"Emerging data indicate that while the Omicron variants enter cells primarily via the TMPRSS2 pathway, they have also developed a notably enhanced dependency on the endosomal cathepsin route."*

Results: Changed the opening description to *"Omicron variants have reduced their dependency on TMPRSS2 but developed a notably enhanced reliance on the endosomal cathepsin route."*

##### **(2) Citation and Discussion:**

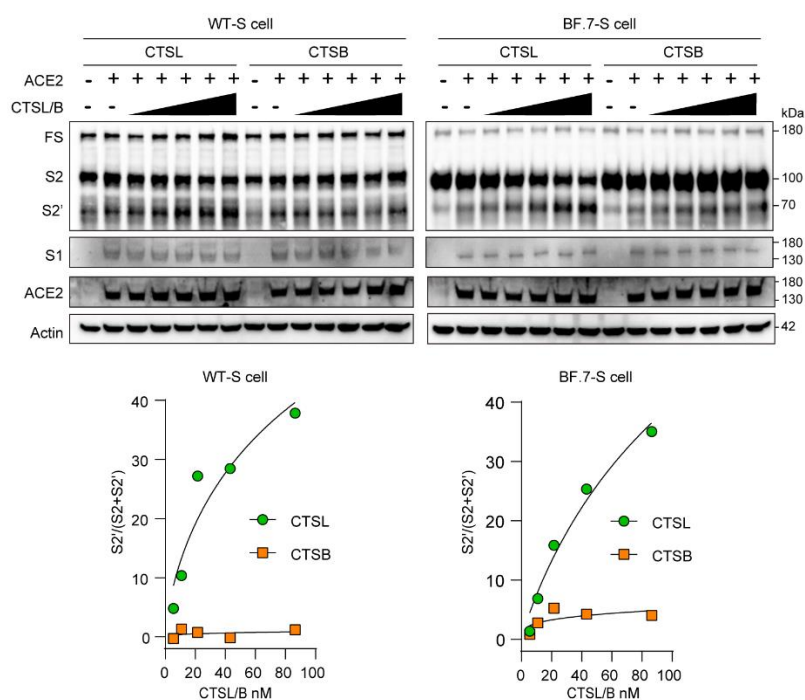
We have cited studies underscoring the continued importance of the TMPRSS2-mediated entry pathway for Omicron (e.g., PMID: 39601592, PMID: 39753670).

Additionally, we have revised the **Discussion** to explicitly note the limitations of our study: “Our cleavage assays and neutralization experiments were performed in defined cell line systems that favor either TMPRSS2- or cathepsin-driven entry, and that our conclusions are restricted to how these pathway biases intersect with S2'-helix accessibility and antibody size. The heterogeneous expression of TMPRSS2 and cathepsins across different tissues leads to variations in viral entry routes. Consequently, the neutralization efficacy of 76E1 against Omicron in vivo is expected to vary by tissue”.

We believe these revisions address the reviewer's concerns by providing a more balanced and precise interpretation of the entry pathway change, firmly grounded in both our data and the broader literature. We thank the reviewer again for this constructive feedback, which has helped strengthen the manuscript.

#### Additional new experiment inspired by this reviewer:

Moreover, the CTSB inhibitor data in the paper PMID: 37769654 that recommended by the reviewer directly prompted us to compare the specific roles of cathepsins in spike cleavage. As the existing literature lacks a direct comparison of CTSL and CTSB proteolytic activity at the S2' site, we undertook a dedicated experiment to resolve this question. This experiment revealed that, across the concentrations tested, CTSB showed minimal to no S2'-site cleavage of both WT or BF.7 spikes, while CTSL mediated robust cleavage (evident by increased S2' and decreased S2 band intensities). These results indicate that CTSL plays a critical role in this context. Therefore, in the **Main text** we present only the 76E1 inhibition data for CTSL cleavage; the comparative CTSB/CTSL cleavage data are provided in **Extended Data Fig. 7**. We are grateful to the reviewer not only for helping us strengthen the manuscript, but also for inspiring this additional line of investigation.



**Extended Data Fig. 7 Comparison of S2'-cleavage efficiency between CTSL and CTSB.** Western blot analysis of dose-dependent S2' cleavage at cell-surface WT or BF.7 spikes by

CTSL or CTSB, with or without ACE2-Fc priming. Cleavage EC<sub>50</sub> values were calculated from band-intensity ratios  $S2' / (S2' + S2)$ . Results demonstrate markedly stronger cleavage by CTSL compared to CTSB across the concentration range tested. Data are representative of  $\geq 2$  independent experiments. Uncropped gel images are shown in Supplementary Fig. 1.

2. The role of H655Y as being responsible for the \*apparent\* switch in entry route is debated with other candidate mutations also having supporting data.

**Response:**

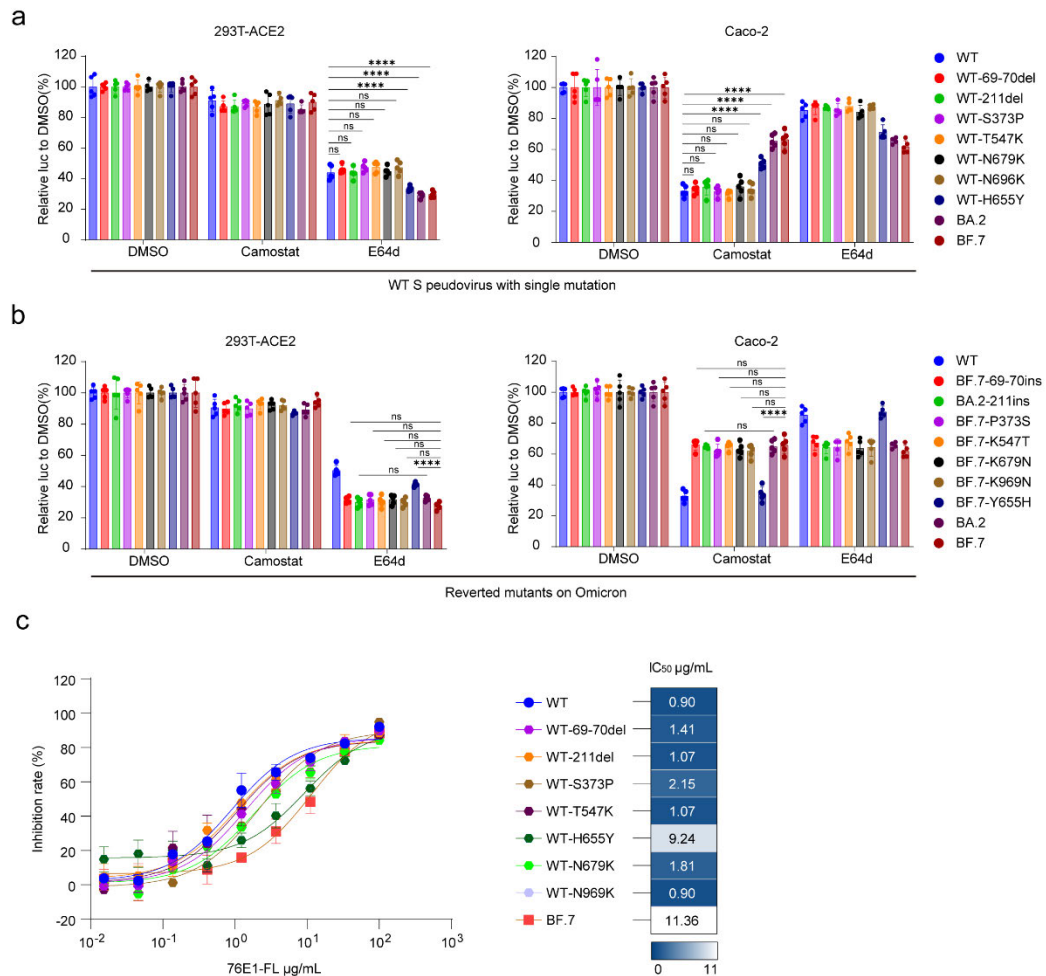
We sincerely thank the reviewer for raising this important point regarding the role of the H655Y mutation and other potential contributors to the altered entry route of Omicron. This feedback has been invaluable in helping us strengthen our manuscript.

Upon a thorough review of the literature, we identified several key studies investigating mutations that influence viral entry pathways. Notably, **PMID: 36039901** identified H655Y as the primary mutation responsible for the shift in Omicron's entry route, while also noting several other mutations (S373P, N679K, 211del, and 60-70del) with modest effects. Similarly, **PMID: 36625591** confirmed the dominant role of H655Y, with T547K exhibiting a minor influence. The **pre-print (bioRxiv <https://www.biorxiv.org/content/10.1101/2022.03.21.485084v1.full.pdf>)** also strongly supports H655Y as the key driver for this phenotypic change. Another **pre-print study (<https://europepmc.org/article/PPR/PPR438894#S2>)** reported that N969K has a slight effect on the entry pathway. A common conclusion across these studies is that H655Y plays a central role, while other mutations exert minor or context-dependent effects.

To systematically validate these findings, we generated pseudoviruses carrying these individual mutations (S373P, N679K, 211del, 60-70del, T547K, N969K) and evaluated their impact on the viral entry pathway. Our experimental results are consistent with the published literature, confirming that these mutations not significantly enhance the reliance on the endosomal pathway and correspondingly reduce dependency on the TMPRSS2-mediated plasma membrane fusion.

Furthermore, we investigated the neutralization susceptibility of these mutant pseudoviruses to the full-length 76E1 antibody. In comparison to the pronounced effect of H655Y, other mutations showed negligible or minimal impacts. Crucially, the H655Y mutation consistently demonstrated the most substantial effect in evading 76E1 neutralization. Therefore, our study justifiably focuses on elucidating the mechanism by which this predominant mutation, H655Y, facilitates escape from 76E1-like antibodies.

In direct response to the reviewer's comment and to provide a complete picture, we have incorporated the data for all these previously reported mutations with negligible or minimal effects into **Supplementary Fig. 7** of our revised manuscript.



**Supplementary Fig. 7 Comparison of H655Y with other reported mutations that modestly influence viral entry pathway.** Mutations 69–70ins, 211ins, P373S, T547K, K679N, and K969N have been previously reported to subtly affect viral entry route; here their impact on pathway choice and 76E1 neutralization is compared with that of H655Y. **a–b**, Relative infection of pseudoviruses carrying the indicated mutations in WT (**a**) or Omicron-reverted (**b**) backgrounds after treatment with 10  $\mu\text{M}$  Camostat or E64d in 293T-ACE2 or Caco-2 cells. Parental WT, BA.2, or BF.7 pseudoviruses were used as controls according to the strain background of each mutation (211ins present in BA.2; others in BF.7). **c**, Neutralization IC<sub>50</sub> of 76E1-FL against the mutant pseudoviruses in 293T-ACE2 cells. Data are mean  $\pm$  SEM ( $n=3$ ); two-sided unpaired t-tests were used to calculate  $P$  values. \*\*\*\*,  $P \leq 0.0001$ .

3. How confident can the authors be regarding “S2'-helix itself is displaced downward by  $\sim 1.2 \text{ \AA}$ ” with a structure determined at worse than  $4 \text{ \AA}$  resolution?

#### Response:

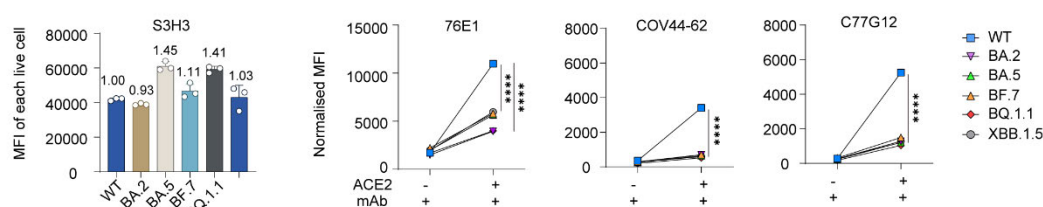
Thanks for pointing this out. We fully appreciate the concern regarding the reliability of quantitative statements at medium resolution. After aligning the two structures modeled by the locally refined maps of WT-S2-76E1 ( $3.73 \text{ \AA}$ ) and XBB.1.5-S2-76E1 ( $4.14 \text{ \AA}$ ) using the CH-CD in S2, we observe a slight displacement of the S2'-helix of approximately  $1.2 \text{ \AA}$ . Given the resolution limit, we have revised the text to describe this change more

conservatively as “the S2'-helix moves downward by ~1 Å”, and other related structural numerical descriptions have also been adjusted accordingly in the revised manuscript.

4. This reviewer suggests using and showing all biological replicates and carrying out statistical analysis for data shown in Fig1b-c,f and 4c-d.

#### Response:

We thank the reviewer for this suggestion. We have now updated the figures as requested. Regarding the figures related to IC<sub>50</sub> values, we have provided the individual IC<sub>50</sub> values from two-three independent biological replicates in the **Supplementary Table 3-6**. In the **Main Text**, the data are presented as the mean values derived from these replicates. For Fig. 1c, replicates were shown and statistical analysis has been performed and is indicated in the figure. For all data, source data are presented in the **Supplementary Information file** and **Source data file**.



**Fig.1c, Flow cytometry analysis of full-length 76E1 binding to cell-surface S variants with or without ACE2-Fc pre-incubation.** Left: relative MFI of S3H3 (comparable neutralization against SARS-CoV-2 variants) to WT S for S expression normalization. Right: normalized MFI of 76E1, COV44-62 and C77G12 relative to S3H3 ( $n=3$ ). Two-sided unpaired t-tests was conducted for the  $P$  values. \*\*\*\*,  $P \leq 0.0001$ .

**Supplementary Table. 3 Summary of pseudovirus neutralizing IC<sub>50</sub> values from three biological replicates. Related to Fig. 1b.**

| IC <sub>50</sub> (μg/mL) |            | WT         | Delta     | BA.2       | BA.5       | BF.7       | BQ.1.1     | XBB.1.5    |
|--------------------------|------------|------------|-----------|------------|------------|------------|------------|------------|
| 76E1                     | Exp 1      | 0.32       | 1.45      | 7.89       | 9.44       | 19.47      | 11.38      | 15.54      |
|                          | Exp 2      | 0.65       | 1.06      | 5.34       | 12.62      | 20.32      | 17.91      | 16.04      |
|                          | Exp 3      | 0.77       | 1.02      | 7.67       | 9.07       | 16.16      | 11.49      | 15.70      |
|                          | Mean ± SEM | 0.58±0.14  | 1.17±0.14 | 6.96±0.82  | 10.38±1.13 | 18.65±1.27 | 13.59±2.16 | 15.76±0.15 |
|                          |            |            |           |            |            |            |            |            |
| COV44-62                 | Exp 1      | 4.48       | 5.95      | 47.43      | 23.70      | 52.65      | 41.52      | 56.17      |
|                          | Exp 2      | 6.02       | 6.22      | 37.12      | 20.68      | 43.79      | 53.34      | 59.08      |
|                          | Exp 3      | 7.15       | 8.64      | 40.48      | 17.27      | 41.48      | 37.03      | 45.90      |
|                          | Mean ± SEM | 5.88±0.77  | 6.94±0.85 | 41.68±3.04 | 20.55±1.86 | 45.97±3.40 | 43.96±4.86 | 53.72±4.00 |
|                          |            |            |           |            |            |            |            |            |
| C77G12                   | Exp 1      | 11.25      | 7.48      | 31.53      | 31.33      | 52.61      | 39.09      | 62.69      |
|                          | Exp 2      | 10.73      | 5.27      | 19.25      | 26.77      | 43.42      | 48.99      | 62.44      |
|                          | Exp 3      | 8.30       | 7.16      | 25.40      | 27.91      | 47.50      | 54.02      | 57.57      |
|                          | Mean ± SEM | 10.09±0.91 | 6.64±0.69 | 25.39±3.54 | 28.67±1.37 | 47.84±2.66 | 47.37±4.39 | 60.90±1.67 |
|                          |            |            |           |            |            |            |            |            |

**Supplementary Table. 4 Summary of pseudovirus neutralizing IC50 values from two biological replicates. Related to Fig. 1g.**

| IC <sub>50</sub> (nM) |       |         | WT    | Delta | BA.2   | BA.5   | BF.7   | BQ.1.1 | XBB.1.5 |
|-----------------------|-------|---------|-------|-------|--------|--------|--------|--------|---------|
| 76E1                  | Exp 1 | FL      | 5.59  | 7.09  | 65.77  | 72.85  | 62.06  | 161.10 | 115.10  |
|                       |       | Fab     | 6.20  | 7.62  | 47.75  | 20.89  | 28.04  | 104.30 | 63.59   |
|                       |       | scFv    | 5.23  | 7.02  | 21.34  | 10.12  | 7.67   | 28.33  | 18.55   |
|                       | Exp 2 | FL      | 2.50  | 8.41  | 66.64  | 60.68  | 55.37  | 162.50 | 97.43   |
|                       |       | Fab     | 5.08  | 8.48  | 34.44  | 21.69  | 31.99  | 72.40  | 50.47   |
|                       |       | scFv    | 2.27  | 5.87  | 13.73  | 10.13  | 10.77  | 21.10  | 16.76   |
|                       | Mean  | FL      | 4.05  | 7.75  | 66.21  | 66.77  | 58.72  | 161.80 | 106.27  |
|                       |       | Fab     | 5.64  | 8.05  | 41.10  | 21.29  | 30.02  | 88.35  | 57.03   |
|                       |       | scFv    | 3.75  | 6.44  | 17.54  | 10.13  | 9.22   | 24.72  | 17.66   |
|                       |       | FL/Fab  | 0.72  | 0.96  | 1.61   | 3.14   | 1.96   | 1.83   | 1.86    |
|                       |       | FL/scFv | 1.08  | 1.20  | 3.78   | 6.59   | 6.37   | 6.55   | 6.02    |
| COV44-62              | Exp 1 | FL      | 34.74 | 52.10 | 148.50 | 874.90 | 262.40 | 316.30 | 559.50  |
|                       |       | Fab     | 34.11 | 52.70 | 50.94  | 375.60 | 103.50 | 49.64  | 227.50  |
|                       |       | scFv    | 7.64  | 14.23 | 13.52  | 40.26  | 28.12  | 15.42  | 33.82   |
|                       | Exp 2 | FL      | 33.42 | 41.46 | 87.50  | 756.30 | 226.10 | 321.90 | 633.50  |
|                       |       | Fab     | 36.59 | 41.53 | 51.07  | 237.00 | 90.46  | 56.48  | 218.60  |
|                       |       | scFv    | 9.52  | 11.14 | 8.42   | 43.21  | 34.06  | 19.34  | 43.36   |
|                       | Mean  | FL      | 34.08 | 46.78 | 118.00 | 815.60 | 244.25 | 319.10 | 596.50  |
|                       |       | Fab     | 35.35 | 47.12 | 51.01  | 306.30 | 96.98  | 53.06  | 223.05  |
|                       |       | scFv    | 8.58  | 12.69 | 10.97  | 41.74  | 31.09  | 17.38  | 38.59   |
|                       |       | FL/Fab  | 0.96  | 0.99  | 2.31   | 2.66   | 2.52   | 6.01   | 2.67    |
|                       |       | FL/scFv | 3.97  | 3.69  | 10.76  | 19.54  | 7.86   | 18.36  | 15.46   |
| C77G12                | Exp 1 | FL      | 48.37 | 35.15 | 309.10 | 408.30 | 269.20 | 472.60 | 465.50  |
|                       |       | Fab     | 52.51 | 31.11 | 162.80 | 330.00 | 65.06  | 248.60 | 120.80  |
|                       |       | scFv    | 19.65 | 13.98 | 74.52  | 159.50 | 28.74  | 40.89  | 62.00   |
|                       | Exp 2 | FL      | 47.88 | 31.14 | 294.30 | 402.00 | 289.20 | 421.90 | 465.70  |
|                       |       | Fab     | 41.26 | 32.92 | 129.60 | 301.80 | 107.60 | 121.00 | 247.10  |
|                       |       | scFv    | 19.76 | 15.13 | 63.95  | 133.10 | 47.33  | 43.38  | 51.08   |
|                       | Mean  | FL      | 48.37 | 33.15 | 301.70 | 405.15 | 279.20 | 447.25 | 465.60  |
|                       |       | Fab     | 46.89 | 32.02 | 146.20 | 315.90 | 86.33  | 184.80 | 183.95  |
|                       |       | scFv    | 19.71 | 14.56 | 69.24  | 146.30 | 38.04  | 42.14  | 56.54   |
|                       |       | FL/Fab  | 1.03  | 1.04  | 2.06   | 1.28   | 3.23   | 2.42   | 2.53    |
|                       |       | FL/scFv | 2.45  | 2.28  | 4.36   | 2.77   | 7.34   | 10.61  | 8.23    |

**Supplementary Table. 5 Summary of pseudovirus neutralizing IC<sub>50</sub> values from two biological replicates. Related to Fig. 4d.**

| IC <sub>50</sub> (nM) |       | WT      | WT-H655Y | BF.7  | BF.7-Y655H |       |
|-----------------------|-------|---------|----------|-------|------------|-------|
| 76E1                  | Exp 1 | FL      | 9.25     | 45.18 | 78.08      | 20.90 |
|                       |       | Fab     | 7.19     | 15.37 | 27.78      | 17.03 |
|                       |       | scFv    | 5.68     | 5.09  | 4.11       | 9.97  |
|                       | Exp 2 | FL      | 6.04     | 36.09 | 72.77      | 19.03 |
|                       |       | Fab     | 5.48     | 17.43 | 30.41      | 17.19 |
|                       |       | scFv    | 4.09     | 2.98  | 7.75       | 7.12  |
|                       | Mean  | FL      | 7.65     | 40.64 | 75.43      | 19.97 |
|                       |       | Fab     | 6.33     | 16.40 | 29.10      | 17.11 |
|                       |       | scFv    | 4.89     | 4.03  | 5.93       | 8.54  |
|                       |       | FL/Fab  | 1.21     | 2.48  | 2.59       | 1.17  |
|                       |       | FL/scFv | 1.57     | 10.07 | 12.72      | 2.34  |

**Supplementary Table. 6 Summary of pseudovirus neutralizing IC<sub>50</sub> values from two biological replicates. Related to Fig. 4f.**

| IC <sub>50</sub> (nM) |       |      | WT        | WT-H655Y | BF.7  | BF.7-Y655H |       |
|-----------------------|-------|------|-----------|----------|-------|------------|-------|
| 76E1                  | Exp 1 | FL   | 293T-ACE2 | 5.22     | 52.97 | 93.03      | 22.62 |
|                       |       |      | Caco-2    | 6.70     | 13.21 | 12.44      | 10.20 |
|                       |       | scFv | 293T-ACE2 | 4.77     | 8.92  | 5.50       | 9.88  |
|                       |       |      | Caco-2    | 5.04     | 7.61  | 4.57       | 5.15  |
|                       | Exp 2 | FL   | 293T-ACE2 | 7.11     | 54.55 | 70.80      | 29.03 |
|                       |       |      | Caco-2    | 4.91     | 17.26 | 16.05      | 8.65  |
|                       |       | scFv | 293T-ACE2 | 6.74     | 8.29  | 5.67       | 8.69  |
|                       |       |      | Caco-2    | 5.95     | 8.64  | 2.63       | 5.72  |
|                       | Mean  | FL   | 293T-ACE2 | 6.17     | 53.76 | 81.92      | 25.83 |
|                       |       |      | Caco-2    | 5.81     | 15.24 | 14.25      | 9.43  |
|                       |       | scFv | 293T-ACE2 | 5.75     | 8.60  | 5.59       | 9.28  |
|                       |       |      | Caco-2    | 5.50     | 8.13  | 3.60       | 5.43  |

5. The comparison in Fig3 between soluble CTSL and membrane-anchored TMPRSS2 for 76E1 cleavage inhibition may be affected because TMPRSS2 is in cis instead of trans.

#### Response:

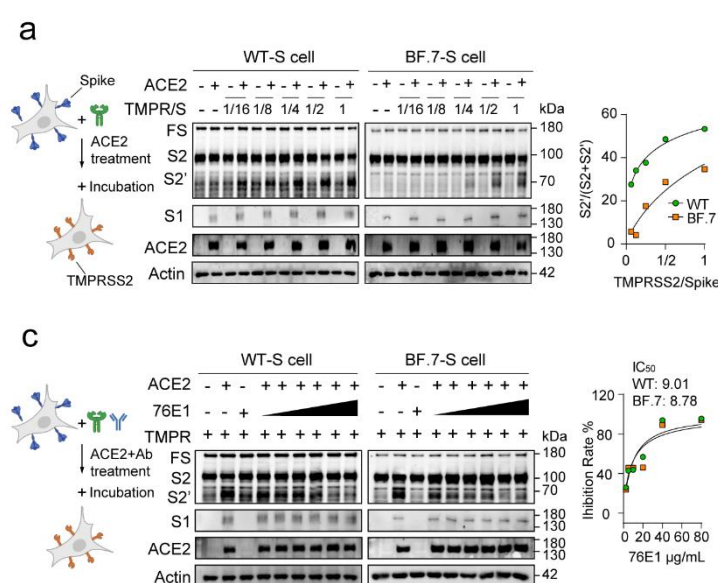
We sincerely thank the reviewer for raising this important methodological point in our original Fig. 3. This is a valuable insight that has directly led to an improvement in the experimental rigor of our study.

In direct response to this comment, we performed a new set of experiments in a trans configuration. Specifically, we co-incubated 293T cells expressing the Spike protein with Vero cells expressing TMPRSS2, followed by treatment with recombinant ACE2 protein. The S2' cleavage efficiency was then assessed by Western Blot and quantified as S2'/(S2'+S2). The results from this trans-cleavage assay were entirely consistent with our

initial cis assay findings, confirming that TMPRSS2 was less efficiently cleaves the Spike protein at the S2' site of Omicron as compared with ancestral strain.

Furthermore, we utilized this trans experimental system to re-evaluate the inhibitory effect of the full-length 76E1 antibody on S2' cleavage. The results again aligned with our previous cis cleavage inhibition data, demonstrating that 76E1-FL effectively inhibits TMPRSS2-mediated cleavage at the S2' site of both Omicron and ancestral strain.

In the revised manuscript, we have replaced the original cis assay data in **Fig. 3** with the results from these newly conducted trans-cleavage and trans-inhibition assays. Moreover, we also have revised the related descriptions in the **Method** section. We believe this revision provides a more physiologically relevant and technically robust comparison, and we are grateful to the reviewer for their suggestion which has strengthened this part of our work.

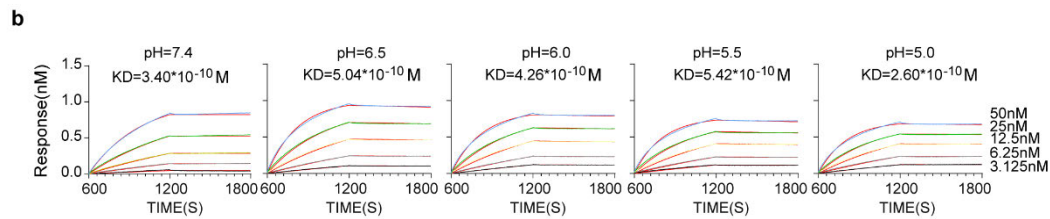


**Fig.3 a**, Dose-dependent S2' trans cleavage of cell-surface WT or BF.7 S by TMPRSS2 assessed by western blot, with or without ACE2 priming. Cleavage activities were derived from S2'/(S2'+S2) band intensity ratios. **c**, Dose-dependent inhibition of TMPRSS2-mediated S2' trans cleavage by 76E1-FL. IC<sub>50</sub> values were derived from S2' band intensities normalized to no-antibody controls.

6. Extended data fig8b should show fit to the data.

## Response:

We thank the reviewer for this suggestion. We have shown the fit to the data in **Extended Data Fig. 8b** as recommended, and the revised figure is included in the manuscript. The fit curves were presented in red color. The fit appropriately models the underlying data, and we believe this addition has strengthened the figure.

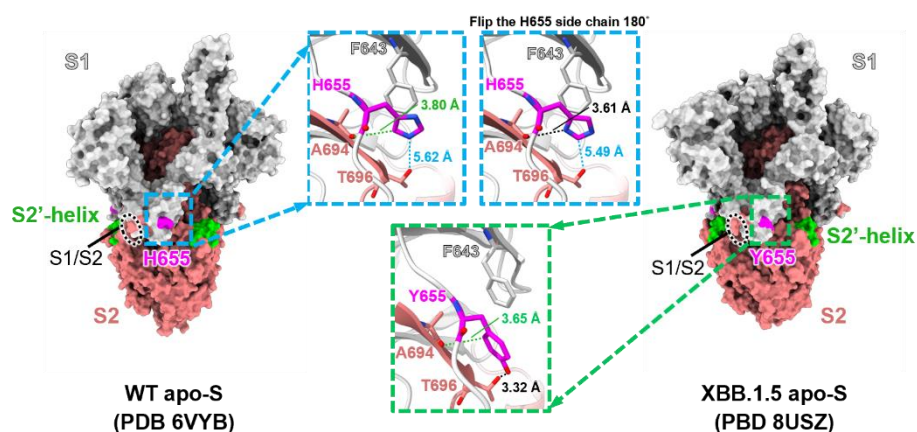


**Extended Data Fig. 8b**, Binding affinity of 76E1-FL antibody to Omicron BF.7 SECD protein as measured by BLI under different low pH conditions. pH 7.4 was included as the control. The curves were fit in red color.

7. This reviewer agrees that a hydrogen bond is formed between S1 Y655 and S2 T696 in XBB.1.5. However, the hydrogen bond formed between the H655 side chain and A694 in WT S (which the authors presumably referred to, although it requires flipping the H655 side chain 180° in the deposited PDB 6VYB structure) is lost in XBB.1.5.

#### Response:

We thank the reviewer for this insightful structural clarification. We agree with the reviewer's comment that, in the deposited prefusion WT S structure (PDB 6VYB), the hydrogen bonds formed between the H655 side chain and the main chain of A694 when flipping the H655 side chain by 180°. Accordingly, we have reorganized structural details in **Fig. 4g** and rephrased the related text to “*Interaction comparison reveals that in prefusion WT S (PDB 6VYB), S1 H655 interacts with the main chain of S2 A694 by van der Waals forces or hydrogen bonds when flipping the H655 side chain by 180°, and with S1 F643 through phenyl ring stacking. Similar interactions are observed in prefusion XBB.1.5 S (PDB 8USZ), where S1 Y655 engages S2 A694 through van der Waals interactions and interacts with S1 F643 via aromatic stacking*” in the revised manuscript.



**Fig. 4g**, Surface representation of prefusion WT (PDB 6VYB, left panel) and XBB.1.5 (PDB 8USZ, right panel) apo-S. The interactions involving H655 are depicted as ribbon sticks in the middle panel. S1 and S2 are colored in light gray and light coral, respectively, with S2'-helix (lime) and H655 or Y655 (magenta) highlighted. The hydrogen bonds are indicated by a black dotted line, and van der Waals forces in green. The blue dashed lines show the distance between specific atoms.

8. Which cell line was used for the HCoV-229E pseudo-virus assay shown in Fig5? Also, it has been established that Vero cells (is it E6, CCL81, another one?) is not a relevant cell line given that it does not faithfully reflect the entry pathway of (at least several strains of) SARS-CoV-2 and leads to loss of the S1/S2 furin cleavage site when used to passage SARS-CoV-2?

**Response:**

We thank the reviewer for raising this important point regarding cell line relevance. We conducted the HCoV-229E pseudo-virus assay in Huh-7 cell line as shown in Fig.5. To the best of our knowledge, there are no established concerns regarding the Huh-7 cell line inaccurately reflecting the entry pathway for HCoV-229E or leading to alterations in viral spike protein cleavage.

Regarding cell lines used for SARS-CoV-2 entry pathway studies, our core investigations into viral entry (Figs. 3-4) were conducted using two distinct and widely adopted cell models: 293T-ACE2 cells, which predominantly favor the endosomal/cathepsin-dependent entry route; Caco-2 cells, which utilize both TMPRSS2-dependent plasma membrane fusion and endosomal/cathepsin-dependent pathways, with the TMPRSS2 route playing a major role.

9. Comparable binding of 76E1 scFv to SARS-CoV and HCoV-229E with vs without receptor in Fig5d is unexpected (the prefusion structure of these spikes also mask the S2' helix).

**Response:**

We thank the reviewer for raising this interesting point regarding the exposure of S2'-helix in our study.

The reviewer rightly notes that the S2'-helix is expected to be shielded in the prefusion conformation of spike proteins. However, our observations align with emerging evidence suggesting coronavirus-specific differences in S2' exposure mechanisms. As reported in the supplementary data of a Nature article (PMID: 35857703), S2'-helix exposure in SARS-CoV-2, SARS-CoV, and MERS-CoV is receptor-dependent, whereas HCoV-229E exhibits receptor-independent S2'-helix exposure. This distinction is consistent with our finding that 76E1 scFv binds similarly to cell-surface HCoV-229E spike regardless of receptor presence.

Regarding the receptor-independent binding of 76E1 Fab/scFv to SARS-CoV spike (Fig. 5d), several studies have reported the coexistence of prefusion and postfusion conformations on mature virions, with their relative ratios potentially vary (PMID: 32694201, PMID: 32979942). Therefore, PMID: 32694201 proposed a model of receptor-independent spike activation. These findings suggest that the spike protein is not strictly confined to the prefusion state but rather exists in a dynamic equilibrium in the viral surface. The occasional exposure of the S2'-helix may mediate a transition to postfusion conformations even in the absence of receptor binding.

This model is supported by our experimental observations: The Fab fragment exhibits only weak binding to unstimulated HCoV-229E spike, consistent with substantial S2' shielding in the predominant prefusion state. In contrast, the minimized scFv shows enhanced binding, suggesting that transient conformational "breathing" creates a sterically

constrained environment that preferentially accommodates smaller binders (**Fig. 5d**).

A parallel phenomenon appears operative for SARS-CoV spike: while receptor binding substantially enhances S2' exposure, we consistently detect low-level binding of both 76E1 Fab even without ACE2. The superior binding capacity of the smaller scFv further reinforces the concept of steric selection during transient spike opening (**Fig. 5d**).

In this study, we have structurally characterized 76E1's interaction with SARS-CoV-2 and Omicron spike intermediate states. While structural analysis of other coronavirus spikes falls beyond the current scope, we fully agree that the mechanistic variations in S2' exposure across different coronaviruses represent a compelling direction for future structural and biophysical investigations.

In light of the reviewer's suggestion, we have put a simple discuss regarding the exposure of S2'-helix in the **Discussion** section to explicitly note the limitations of our study and inform future study: *“Size minimization improves antibody efficacy against Omicron, SARS-CoV-1, and HCoV-229E, but not MERS-CoV or HCoV-NL63, indicating divergent accessibility of the S2'-helix epitope. The underlying exposure mechanisms likely vary among coronaviruses and remain an important area for future research”* .

10. Lines 390-398 should be removed altogether because the cryoEM data are of insufficient quality to speculate on what that undefined class is.

**Response:**

We thank the reviewer for this suggestion. We agree that speculative data should not be included in a scientific manuscript. In response, we have removed the relevant descriptions in the text (Lines 390-398 in previous version) and the figure (Supplementary Fig. 4 in previous version) from the revised manuscript.

**Referee #4 (Remarks to the Author):**

The authors have undertaken significant efforts to address my concerns, as well as those raised by the other reviewers. I am satisfied with the revisions and believe they satisfactorily resolve the issues I previously identified.

**Response:**

We sincerely appreciate the positive comments on our revised manuscript.